

Stylized Facts on Multinational Firms and Trade in Latin America

Each fact with its original exhibit reproduced on the current base and the variants for the working paper — all goods

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1 How to read this document

The document follows the six stylized facts of the July-2026 note (v. 2026-07-29). Each section opens with the fact, reproduces its *original* exhibit on the current base, and then places under it every variant prepared for the working paper: the figures with the foreign-MNE bar split by the multinational’s home country (item 1a); alternative product-sophistication measures for Figure 2, including the Fontagné–Guimbard–Orefice (2022) elasticities, with the full distributions (1b); two-way tables and heat maps of parent country $\times$ destination (1c, 1d); the decomposition of multinationals into foreign vs domestic and of foreign MNEs into present-through-headquarters vs not, in the presence and distance regressions (1e); products at HS6 with descriptions (1f). The four sectors are summarised at the end; the sector-by-sector version of every exhibit is the companion document *WP_sectors*, and the country-by-country exhibits are in *WP_countries*. Everything is generated by `stylized_facts/python/wp_*.py` in the project repository (https://github.com/jsvp97/IADB_Orbis_DNB_Customs_Final); the reader’s guide is `docs/WP_RESULTS_2026-09-07.md`.

Conventions (identical to the note). Matched = the firm was found in Orbis or D&B (`_merge_DNB_Orbis=3`); domestic MNE = parent in the exporting country; foreign MNE = matched minus domestic, so matched firms with no recorded parent count as foreign; nine origins (Ecuador excluded); value-weighted; pooled 2006–2022; navy = foreign, light gray = domestic. Wherever the foreign bar is split by home country, the ten largest parent countries (by foreign-MNE export value in the scope) are shown in shades of navy (rank 1 darkest) and every other foreign MNE—including those with no recorded parent—is pooled into “Other foreign MNEs” (gray-blue); domestic MNEs keep the light gray. Two-way country tables show the top-10 parents and destinations; heat maps the top 15. Regressions use plain logs (a zero count drops the cell from that column; no $\ln(1+x)$) and cluster at origin–destination. Sectors: agriculture = HS 01–24; mining & fuels = HS 25–27 and 71; manufacturing = HS 28–97 excluding 71; rest = HS 98–99, chapter 00 and unclassifiable codes (services are not in customs data).

Same data as the note. The regressions that depend only on the matched flag reproduce the note to the last digit: Table 20 returns the note’s Table 1 Panel A coefficients and every observation count (2.0823 / 2.0685 / 0.9414; extensive 1.4622 / 1.4591 / 0.6818; $N = 1,008,472 \dots 1,137,550$) and Table 24 the note’s Table 2 pooled coefficients (−0.1639 / −0.1848; 0.0461 / 0.0458; $N = 5,081,547 / 5,081,360$); Figure 19 gives the note’s 56 % of value for parents with more than 100 affiliates. The cells that use the group’s affiliate links differ in the third decimal only (Table 1 Panel B 1.3536 vs the note’s 1.3522; Table 2 “present” 0.0689 vs 0.0673), because the current base carries the affiliate and parent links of the 2026-04-21 build. The base is therefore the same 5.45 M firm–destination–product–year rows. The field that differs materially is the *parent country*: the current base carries a recorded parent for 92 % of foreign-MNE export value (script 02 and the AI review of unknown parents, merged in the 2026-04-21 build), whereas the note’s Figure 4 was drawn on a copy in which about half of that value had no recorded parent. The parent ranking becomes USA > GBR > CAN > LIE > DEU > NLD (Liechtenstein: foundations owning Argentine and Chilean exporters of copper ore, oil-cake and cereals—a conduit), and the domestic-MNE share rises where the recovered parents are the origin country (COL 0.28 $\rightarrow$ 0.39, CHL 0.02 $\rightarrow$ 0.14), so Fact 1’s “foreign $\gg$ domestic” needs a qualification for Colombia.

Two data limits. (i) “Headquarters exporters” are not identifiable: every matched exporter has a global ultimate owner different from itself (0 of 5.45 M rows), because the match runs through ownership links. What is identifiable is the group’s presence *at the destination* through its headquarters (the affiliate ships to the parent’s country); Facts 5 and 6 decompose foreign MNEs on that. (ii) On the substitution elasticities: Fontagné, Guimbard and Orefice (2022) estimate one import-demand elasticity per HS6 product (rev. 2007), common to all exporters and importers—the elasticity is identified from tariff variation across the universe of bilateral flows of the product and is a characteristic of the product, not of the origin. Assigning the same $|\sigma|$ to a product whatever its exporting country is therefore the intended use of the database (non-significant or positive HS6 estimates are replaced by the source with the HS4 average and flagged); no origin-specific

adjustment is needed or possible. The Rauch (1999) classification (conservative, SITC 3/4 digits mapped to HS6) is added; the Micro-D classification of Bernini, González, Hallak and Viccondoa (2018) is defined on Argentina's 12-digit nomenclature from package-size attributes and cannot be carried to HS6 data for other countries.

2 Fact 1 — Multinational corporations account for a large share of export values across countries

2.1 Original exhibit

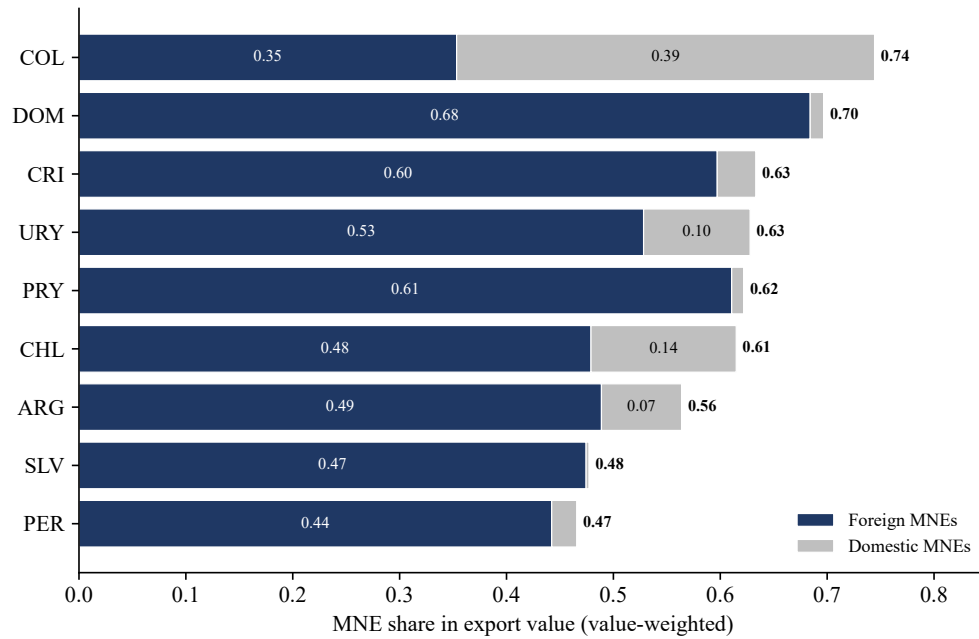


Figure 1: Figure 1 of the note, reproduced on the current base: MNE share of export value by origin, foreign vs domestic.

Note: Value-weighted; pooled 2006–2022; nine LAC origins. Bar length = total MNE share; numbers in the appendix.

2.2 Variant: the foreign bar split by the multinational's home country (item 1a)

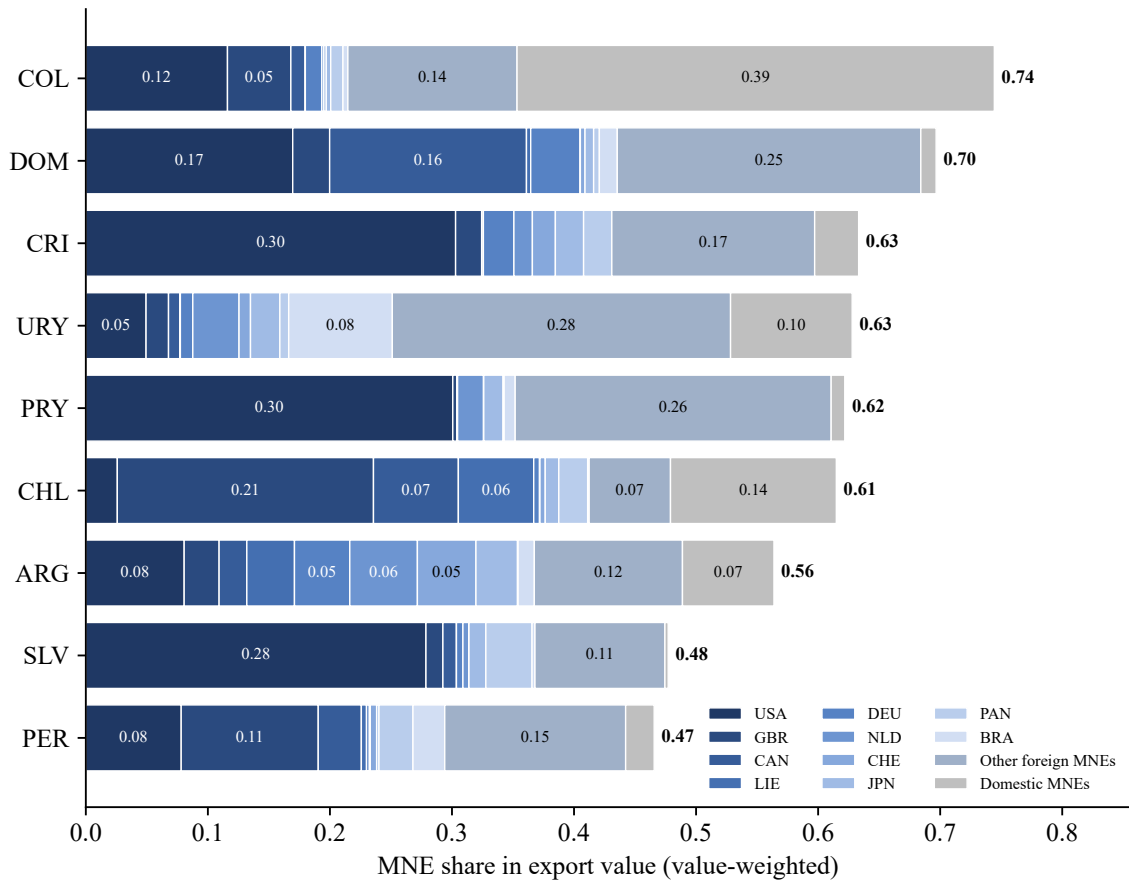


Figure 2: MNE share of export value by origin, foreign bar split by parent country (top 10, Other foreign, Domestic).



Figure 3: companion heat map: share of each origin's export value by parent-country group (%).

3 Fact 2 — Foreign multinationals specialize in complex products, domestic ones in primary goods

3.1 Original exhibits

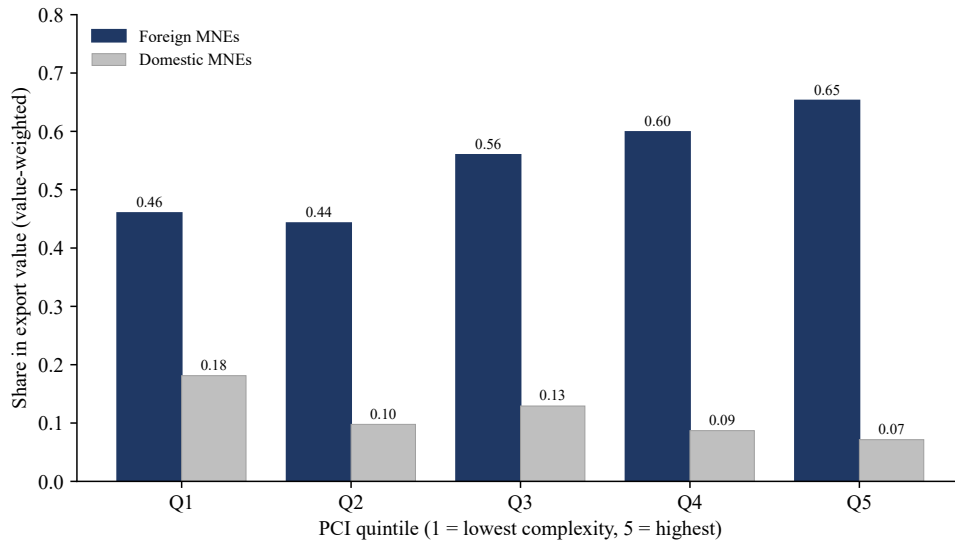


Figure 4: Figure 2 of the note, reproduced: MNE share of export value by Product Complexity Index quintile.

Note: Value-weighted at HS6; quintiles of the Hausmann–Hidalgo PCI over the scope’s HS6 products (Q1 = least complex).

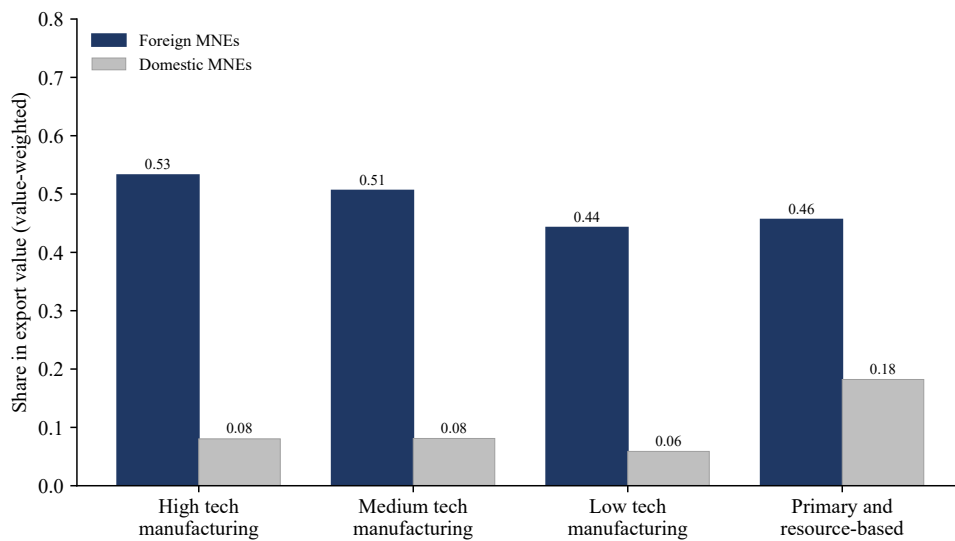


Figure 5: Figure 3 of the note, reproduced: MNE share of export value by Lall (2000) technology category.

3.2 Variant: split by the multinational's home country (item 1a)

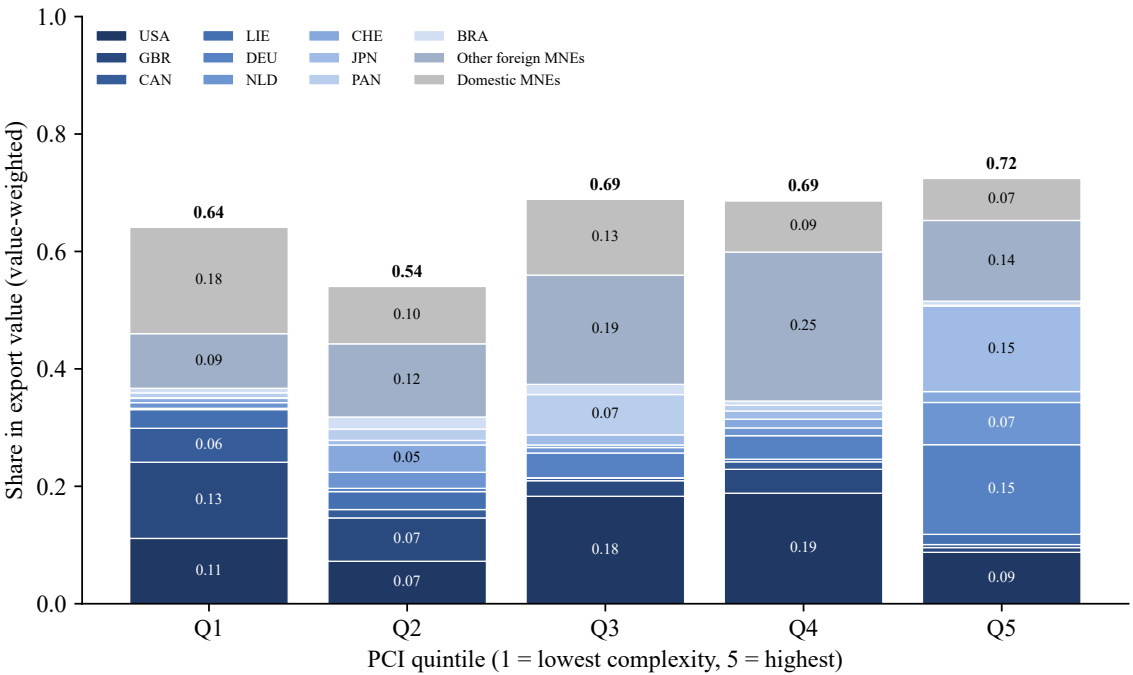


Figure 6: Figure 2 with the foreign bar split by parent country.

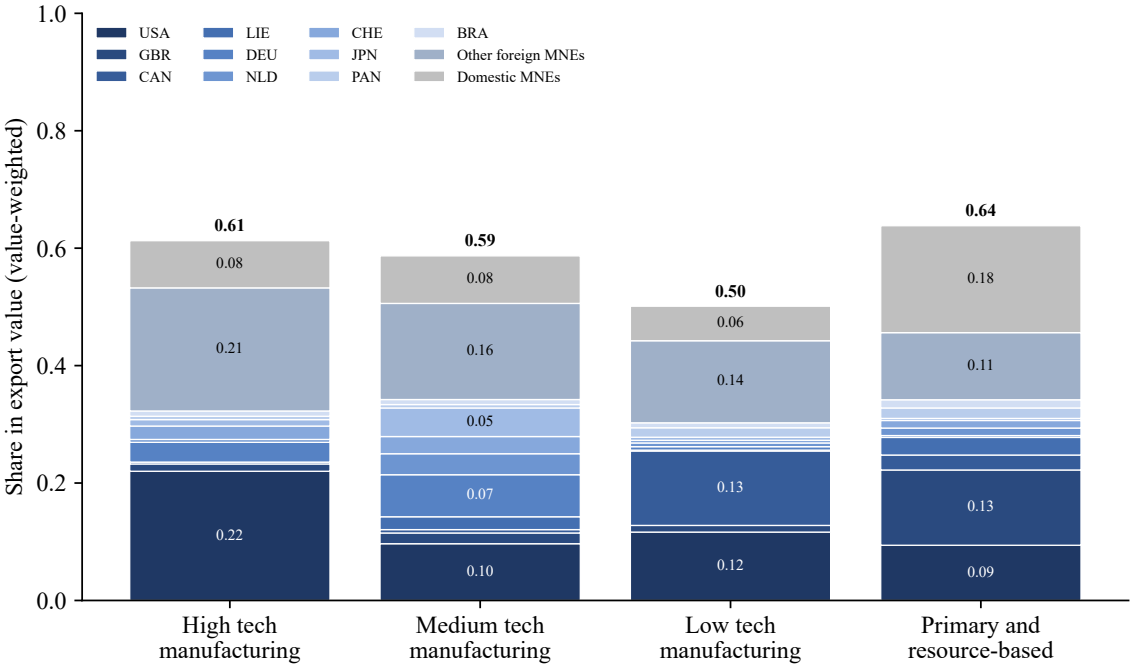


Figure 7: Figure 3 with the foreign bar split by parent country.

3.3 Variant: other measures of product sophistication; substitution elasticities (item 1b)

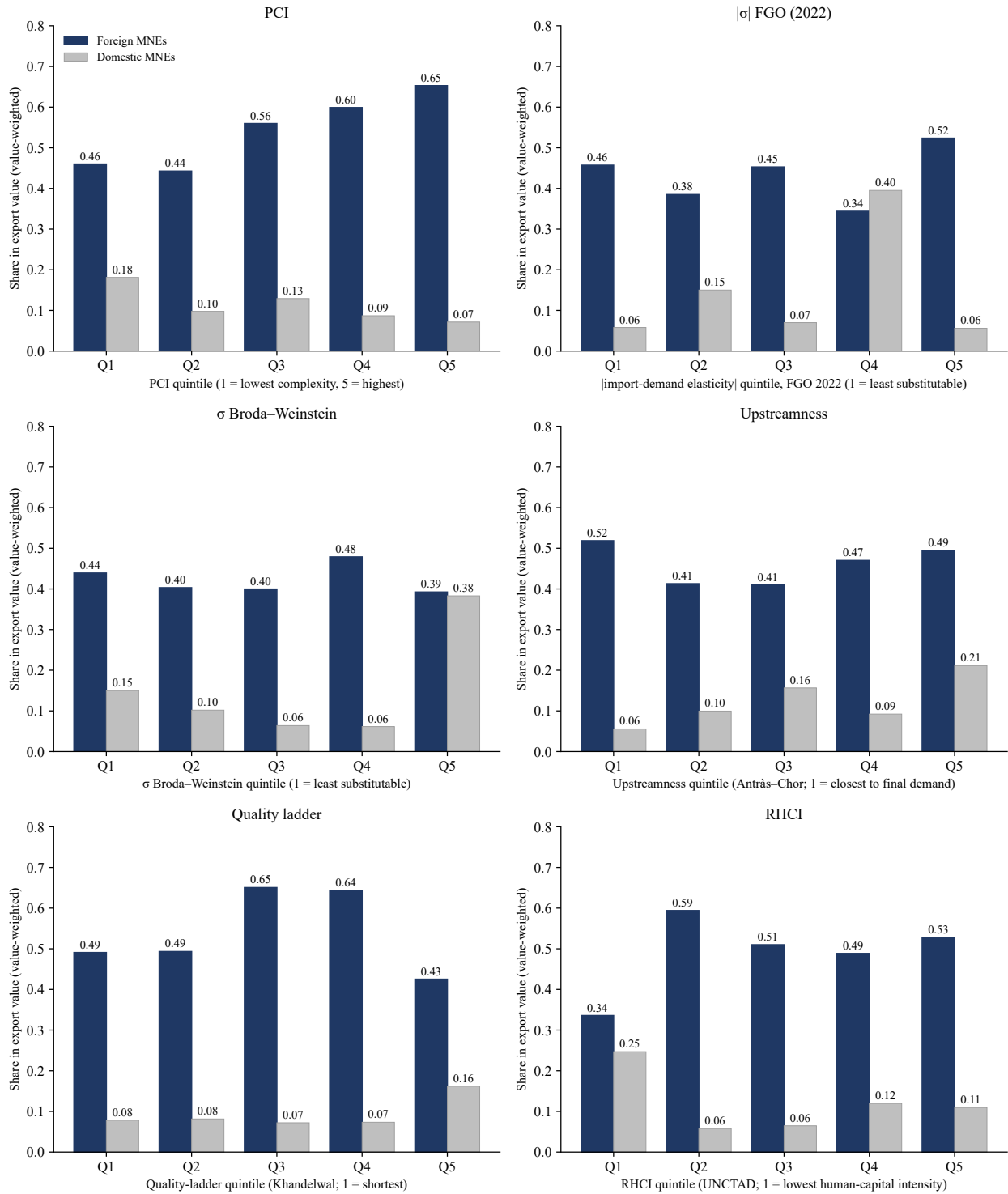


Figure 8: Figure 2 redrawn for six sophistication measures (foreign navy, domestic gray; quintiles over the scope's HS6 products).

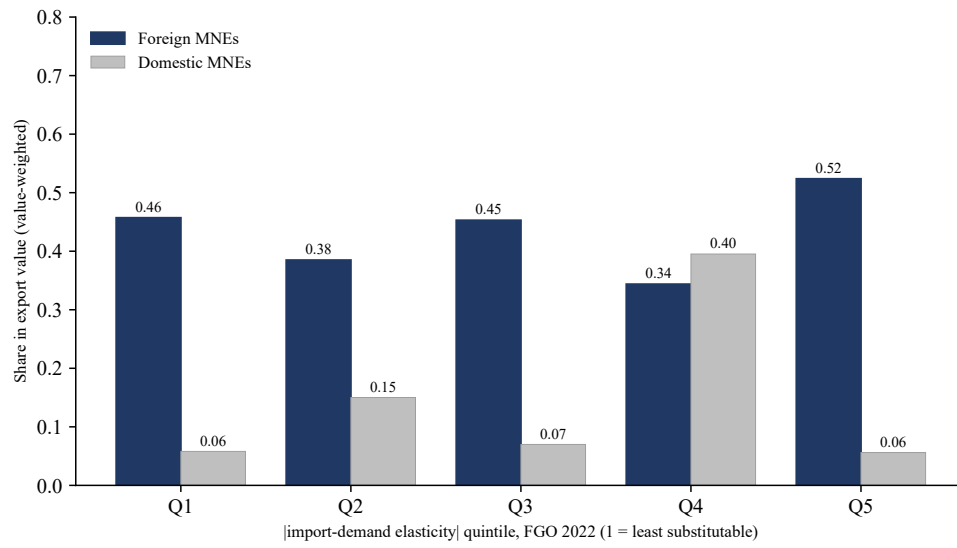


Figure 9: Figure 2 with $|\sigma|$ from Fontagné, Guimbard and Orefice (2022) in place of the PCI.

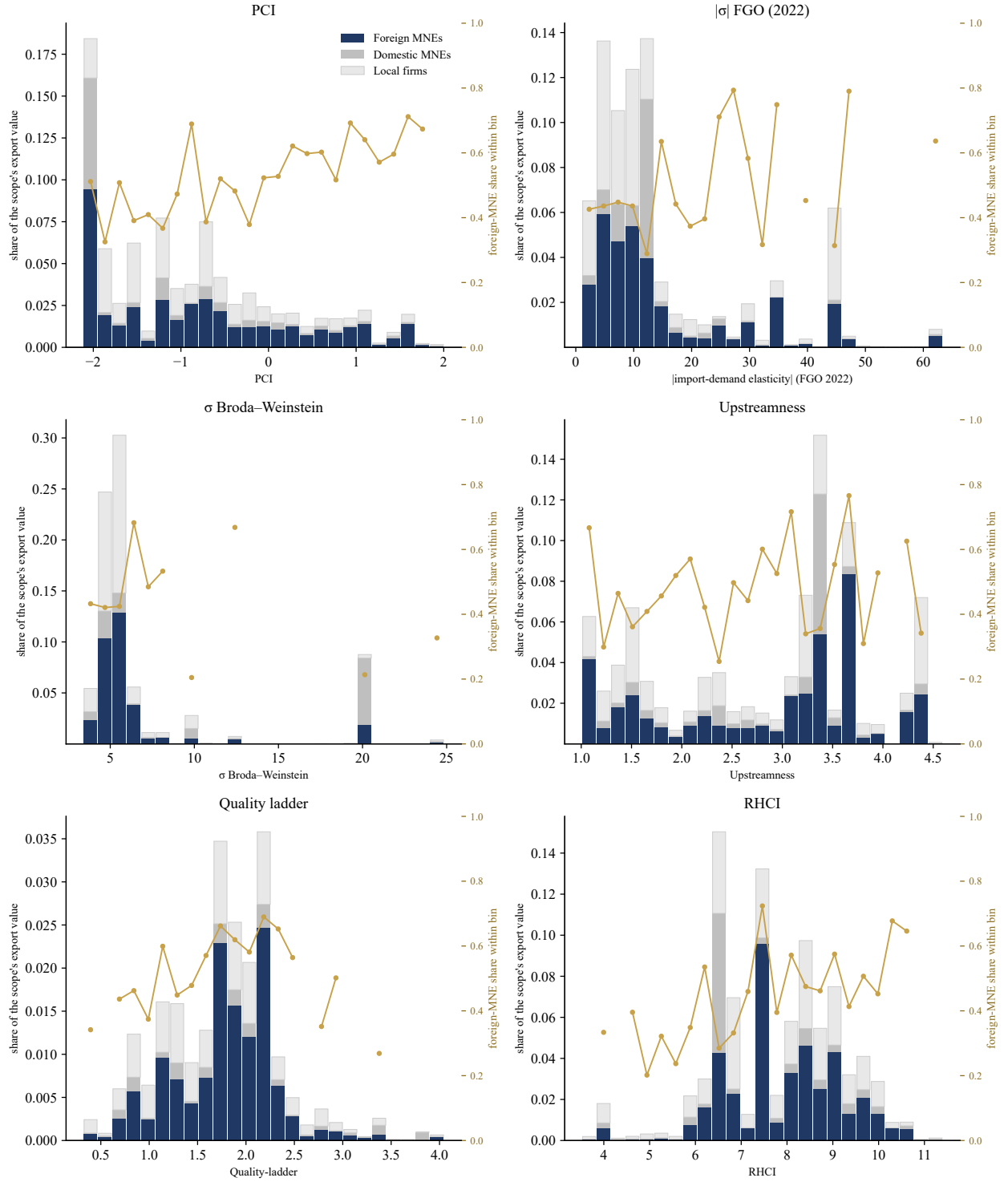


Figure 10: Full distributions: export value across bins of each measure, stacked by owner type, with the foreign-MNE share of each bin (line, right axis).

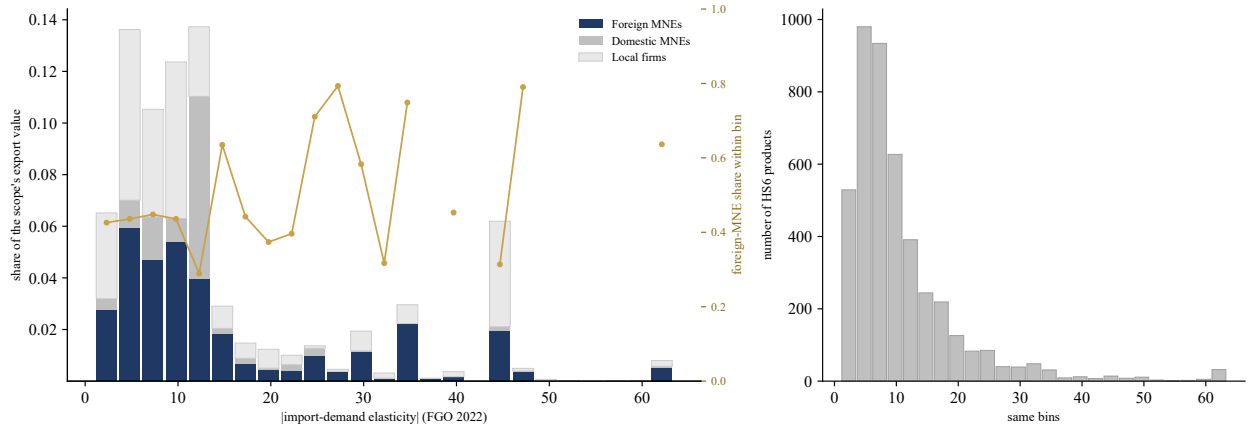


Figure 11: Distribution over $|\sigma|$ (FGO 2022): export value by owner type and foreign share per bin (left); number of HS6 products per bin (right).

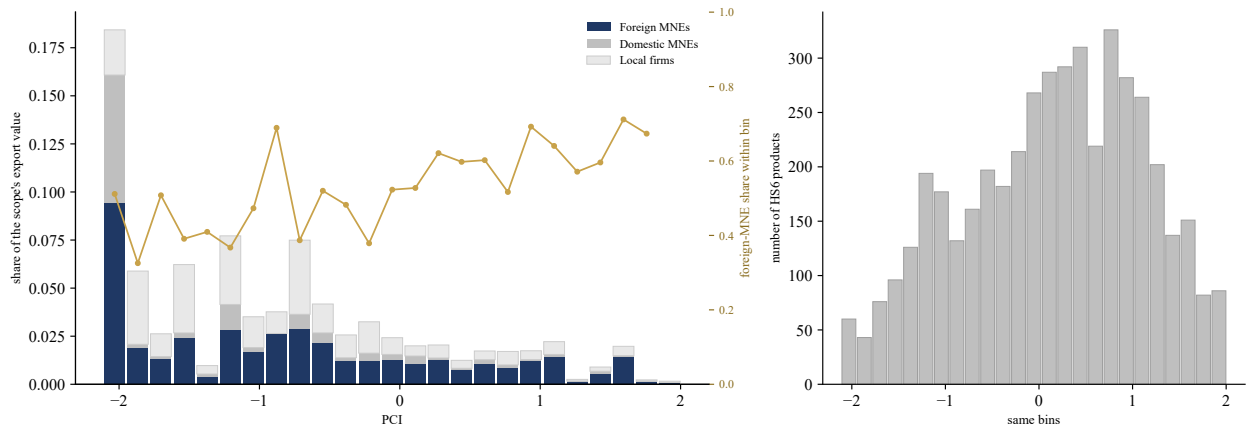


Figure 12: Distribution over the PCI: export value by owner type and foreign share per bin (left); number of HS6 products per bin (right).

Table 1: Foreign and domestic MNE shares by quintile, six sophistication measures

Measure	Foreign MNE share					Domestic MNE share				
	Q1	Q2	Q3	Q4	Q5	Q1	Q2	Q3	Q4	Q5
PCI	0.46	0.44	0.56	0.60	0.65	0.18	0.10	0.13	0.09	0.07
$ \sigma $ FGO (2022)	0.46	0.38	0.45	0.34	0.52	0.06	0.15	0.07	0.40	0.06
σ Broda–Weinstein	0.44	0.40	0.40	0.48	0.39	0.15	0.10	0.06	0.06	0.38
Upstreamness	0.52	0.41	0.41	0.47	0.49	0.06	0.10	0.16	0.09	0.21
Quality ladder	0.49	0.49	0.65	0.64	0.43	0.08	0.08	0.07	0.07	0.16
RHCI	0.34	0.59	0.51	0.49	0.53	0.25	0.06	0.06	0.12	0.11

Value-weighted shares of export value within quintile; quintiles over HS6 products (unweighted); ‘–’ = fewer than 30 HS6 products carry the measure in this scope. $|\sigma|$ FGO: absolute value of the tariff-based import-demand elasticity of Fontagné, Guimbard and Orefice (2022), HS6 rev. 2007 (non-significant or positive HS6 estimates replaced by the HS4 average by the source); larger = more substitutable. σ Broda–Weinstein, upstreamness (Antràs–Chor), quality ladder (Khandelwal), RHCI (UNCTAD) as in the document’s appendix.

Table 2: Correlations across sophistication measures (HS6, value-weighted)

	PCI	$ \sigma $ FGO (2022)	σ Broda–Weinstein	Upstreamness	Quality ladder	RHCI
PCI	1.00	-0.22	-0.49	-0.49	0.28	0.74
$ \sigma $ FGO (2022)	-0.22	1.00	0.03	0.62	0.12	0.02
σ Broda–Weinstein	-0.49	0.03	1.00	0.20	-0.11	-0.25
Upstreamness	-0.49	0.62	0.20	1.00	0.12	-0.19
Quality ladder	0.28	0.12	-0.11	0.12	1.00	0.34
RHCI	0.74	0.02	-0.25	-0.19	0.34	1.00

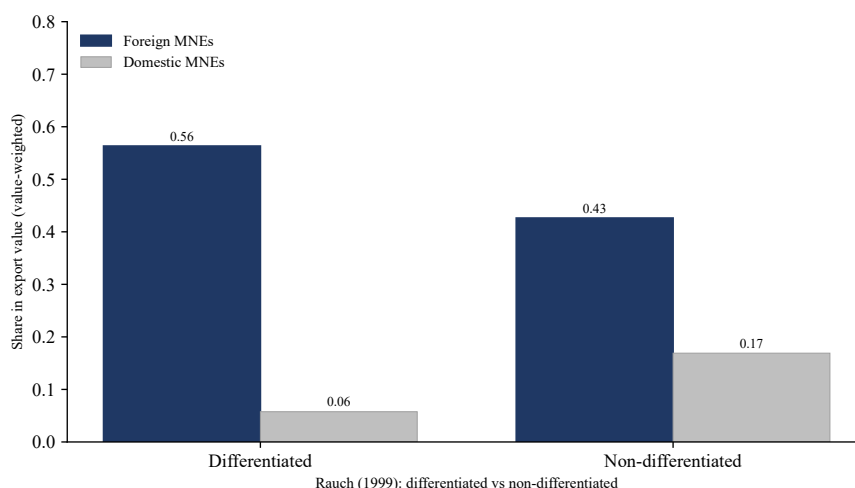
Value-weighted Pearson correlations, pairwise: each cell uses the HS6 products carrying both measures (3,124 to 4,841 products; ‘–’ = fewer than 30).

Table 3: MNE shares by Rauch (1999) class

Rauch (1999) class	Foreign	Domestic	MNE total	N HS6	Value (\$bn)
Differentiated	0.563	0.058	0.621	2,712	450.4
Reference-priced	0.549	0.082	0.631	1,636	698.5
Homogeneous (exchange)	0.343	0.228	0.571	319	1,030.0

Table 4: MNE shares by Rauch (1999) class, two classes: differentiated vs non-differentiated (reference-priced + homogeneous)

Rauch (1999): differentiated vs non-differentiated	Foreign	Domestic	MNE total	N HS6	Value (\$bn)
Differentiated	0.563	0.058	0.621	2,712	450.4
Non-differentiated	0.426	0.169	0.595	1,955	1,728.4

**Figure 13:** Foreign vs domestic MNE shares of export value: differentiated vs non-differentiated products (Rauch 1999; non-differentiated = reference-priced + homogeneous).**Table 5:** MNE shares by BEC end use

BEC end use	Foreign	Domestic	MNE total	N HS6	Value (\$bn)
Intermediate	0.476	0.155	0.631	3,072	1,667.8
Consumption	0.409	0.087	0.496	1,193	449.1
Capital	0.647	0.056	0.703	659	88.5
Mixed (passenger cars)	0.885	0.000	0.885	8	28.9
n.e.s.	0.198	0.345	0.543	20	62.6

Table 6: Note's Table A.4 ladder with the FGO elasticity added (ODPY cells, MNE value share)

	(1)	(2)	(3)
<i>Panel A: MNE_{total} share</i>			
Complexity (PCI)	0.0281*** (0.0047)	0.0307*** (0.0045)	0.0409*** (0.0040)
σ FGO (per s.d.)	-0.0078*** (0.0026)	-0.0074*** (0.0025)	-0.0057*** (0.0021)
Upstreamness	0.0087* (0.0047)	0.0079* (0.0044)	0.0108*** (0.0038)
<i>Panel B: MNE_{ext} share</i>			
Complexity (PCI)	0.0926*** (0.0049)	0.0925*** (0.0047)	0.0855*** (0.0043)
σ FGO (per s.d.)	0.0339*** (0.0023)	0.0331*** (0.0022)	0.0241*** (0.0020)
Upstreamness	-0.0239*** (0.0043)	-0.0253*** (0.0041)	-0.0148*** (0.0034)
<i>Panel C: MNE_{dom} share</i>			
Complexity (PCI)	-0.0645*** (0.0053)	-0.0618*** (0.0048)	-0.0446*** (0.0038)
σ FGO (per s.d.)	-0.0417*** (0.0027)	-0.0405*** (0.0025)	-0.0298*** (0.0022)
Upstreamness	0.0326*** (0.0041)	0.0332*** (0.0037)	0.0256*** (0.0028)
Origin FE	✓		
Destination FE	✓		
Year FE	✓		
Origin $\times$ year FE		✓	
Destination $\times$ year FE		✓	
Origin $\times$ dest. $\times$ year FE			✓
Observations	1,563,005	1,562,745	1,561,136

Origin-destination-product-year cells; dep. var. MNE value share; weighted by total trade value; robust (HC1) SE. | σ | FGO standardised (mean 0, s.d. 1). *** p<0.01, ** p<0.05, * p<0.1

4 Fact 3 — Multinational corporations from a small set of countries dominate exports

4.1 Original exhibit

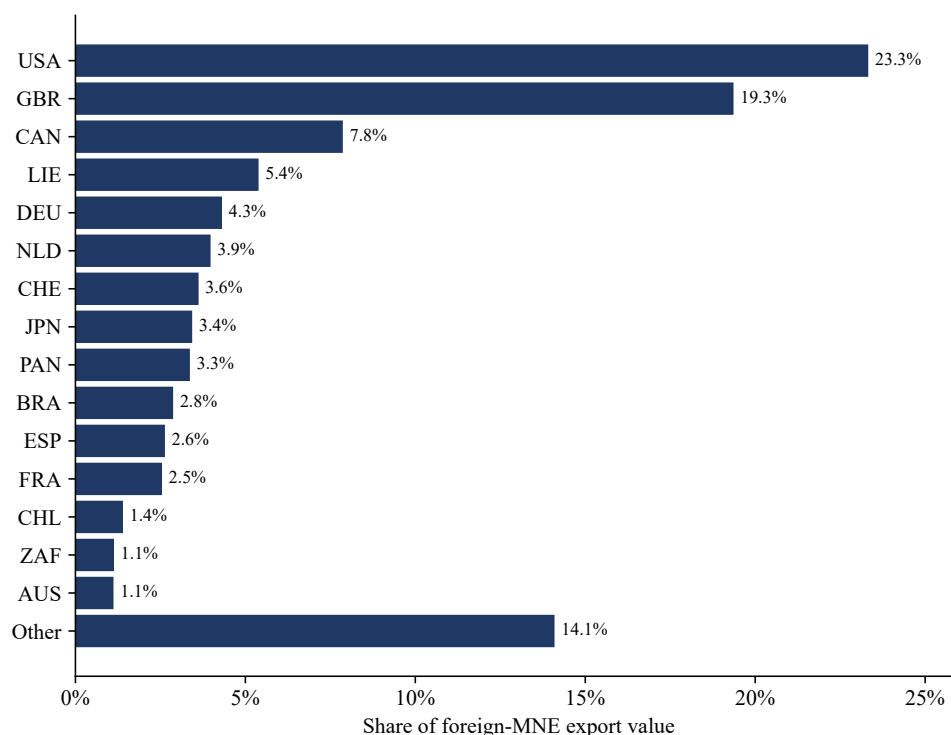


Figure 14: Figure 4 of the note, reproduced on the current base: foreign-MNE export value by parent country, top 15 plus Other.

Note: Foreign MNEs with a recorded parent country.

Table 7: Foreign-MNE export value by parent country (numbers behind the figure)

Parent country	Share (%)	Value (\$bn)
USA	23.3	238.9
GBR	19.3	198.3
CAN	7.8	80.4
LIE	5.4	55.0
DEU	4.3	43.9
NLD	3.9	40.5
CHE	3.6	36.9
JPN	3.4	34.9
PAN	3.3	34.2
BRA	2.8	29.2
ESP	2.6	26.7
FRA	2.5	25.8
CHL	1.4	14.1
ZAF	1.1	11.3
AUS	1.1	11.2
Other (150 countries)	14.1	144.3
Foreign MNEs with recorded parent	100.0	1,025.6
Foreign MNEs, parent unknown (excluded above)	—	83.6

4.2 Variant: parent country × export destination — tables (item 1c)

Table 8: Parent region × destination region, export value (USD bn)

Parent region / Destination region	Latin America	North America	Europe	Asia	Africa	Oceania	Rest of World	Total
Latin America	49.5	19.0	18.1	33.4	1.5	0.9	0.4	122.6
North America	100.0	75.5	63.9	73.7	5.6	0.3	0.3	319.3
Europe	136.8	57.3	87.2	188.2	14.3	5.1	0.6	489.5
Asia	35.1	12.9	8.1	11.7	0.8	0.1	0.2	68.8
Africa	1.5	0.9	3.7	7.4	0.3	0.0	0.0	13.9
Oceania	2.3	0.6	0.9	7.3	0.0	0.0	0.0	11.2
Rest of World	0.1	0.1	0.0	0.0	0.0	0.0	0.0	0.2
Foreign MNE, parent unknown	32.7	18.7	15.6	14.3	1.1	0.4	0.8	83.6
Domestic MNEs	87.6	111.4	37.7	91.5	7.7	1.7	0.9	338.6
Local firms (unmatched)	256.8	204.0	205.3	243.5	27.0	5.7	3.6	946.0
Total	702.4	500.3	440.6	671.1	58.2	14.2	6.9	2,393.8

Export value, USD bn. Pooled 2006–2022, nine LAC origins (Ecuador excluded), all destinations. Parent = country of the multinational's ultimate parent as recorded (Orbis/D&B). Rows for domestic MNEs and local exporters are given for comparison.

Table 9: Parent region × destination region, row %

Parent region / Destination region	Latin America	North America	Europe	Asia	Africa	Oceania	Rest of World	Total
Latin America	40.4	15.5	14.7	27.2	1.2	0.7	0.3	100.0
North America	31.3	23.6	20.0	23.1	1.7	0.1	0.1	100.0
Europe	28.0	11.7	17.8	38.4	2.9	1.0	0.1	100.0
Asia	51.0	18.7	11.7	17.0	1.1	0.2	0.3	100.0
Africa	10.9	6.6	26.9	53.2	2.2	0.0	0.2	100.0
Oceania	20.1	5.7	8.4	65.5	0.1	0.2	0.0	100.0
Rest of World	32.6	25.0	20.9	14.3	0.0	7.1	0.0	100.0
Foreign MNE, parent unknown	39.1	22.4	18.7	17.1	1.3	0.4	1.0	100.0
Domestic MNEs	25.9	32.9	11.1	27.0	2.3	0.5	0.3	100.0
Local firms (unmatched)	27.1	21.6	21.7	25.7	2.9	0.6	0.4	100.0

Row percentages: destination mix of each row group. Pooled 2006–2022, nine LAC origins (Ecuador excluded), all destinations. Parent = country of the multinational's ultimate parent as recorded (Orbis/D&B). Rows for domestic MNEs and local exporters are given for comparison.

Table 10: Parent region × destination region, column %

Parent region / Destination region	Latin America	North America	Europe	Asia	Africa	Oceania	Rest of World
Latin America	7.0	3.8	4.1	5.0	2.5	6.1	5.5
North America	14.2	15.1	14.5	11.0	9.6	1.8	4.3
Europe	19.5	11.4	19.8	28.0	24.5	36.1	9.1
Asia	5.0	2.6	1.8	1.7	1.3	0.9	2.8
Africa	0.2	0.2	0.9	1.1	0.5	0.0	0.4
Oceania	0.3	0.1	0.2	1.1	0.0	0.1	0.0
Rest of World	0.0	0.0	0.0	0.0	0.0	0.1	0.0
Foreign MNE, parent unknown	4.7	3.7	3.5	2.1	1.9	2.6	12.3
Domestic MNEs	12.5	22.3	8.6	13.6	13.3	11.9	13.1
Local firms (unmatched)	36.6	40.8	46.6	36.3	46.4	40.4	52.5
Total	100.0	100.0	100.0	100.0	100.0	100.0	100.0

Column percentages: who supplies each destination. Pooled 2006–2022, nine LAC origins (Ecuador excluded), all destinations. Parent = country of the multinational's ultimate parent as recorded (Orbis/D&B). Rows for domestic MNEs and local exporters are given for comparison.

Table 11: Top-10 parents × top-10 destinations, export value (USD bn)

Parent / Destination	CHN	USA	BRA	JPN	ESP	CHL	KOR	CAN	NLD	ARG	Other destinations	Total
USA	24.1	53.4	18.6	9.0	9.5	6.9	5.0	1.2	8.2	11.7	91.4	238.9
GBR	72.4	11.2	7.7	14.0	5.9	6.7	12.8	6.5	6.4	0.4	54.2	198.3
CAN	10.3	5.3	3.4	4.3	2.5	0.4	1.5	15.7	0.7	0.2	36.2	80.4
LIE	13.7	2.0	2.0	10.3	1.9	1.4	3.0	0.0	1.0	0.0	19.6	55.0
DEU	0.4	6.5	14.0	0.5	0.8	1.4	0.3	0.5	0.4	0.4	18.8	43.9
NLD	3.7	1.6	10.9	0.2	0.4	1.5	0.5	0.1	0.9	1.3	19.6	40.5
CHE	2.3	1.8	2.5	0.2	1.1	1.3	0.3	0.2	0.7	0.2	26.3	36.9
JPN	0.8	3.0	15.7	3.1	0.2	0.5	0.1	0.2	0.3	0.7	10.3	34.9
PAN	12.0	3.3	0.4	0.6	0.6	0.4	2.1	0.2	0.7	0.1	13.9	34.2
BRA	3.8	4.1	5.4	0.5	0.2	0.9	0.2	0.2	0.6	0.7	12.7	29.2
Other parents	22.2	45.6	27.3	5.2	8.6	9.3	3.7	3.8	4.9	8.7	93.9	233.4
Total	165.8	137.8	107.9	47.9	31.6	30.7	29.5	28.5	25.0	24.3	396.9	1,025.6

Export value, USD bn. Foreign MNEs with a recorded parent country; top 10 parents and top 10 destinations by foreign-MNE export value. Pooled 2006–2022, nine LAC origins (Ecuador excluded), all destinations. Parent = country of the multinational's ultimate parent as recorded (Orbis/D&B).

Table 12: Top-10 parents × top-10 destinations, row %

Parent / Destination	CHN	USA	BRA	JPN	ESP	CHL	KOR	CAN	NLD	ARG	Other destinations	Total
USA	10.1	22.3	7.8	3.8	4.0	2.9	2.1	0.5	3.5	4.9	38.2	100.0
GBR	36.5	5.6	3.9	7.1	3.0	3.4	6.5	3.3	3.2	0.2	27.3	100.0
CAN	12.8	6.6	4.2	5.3	3.1	0.5	1.9	19.5	0.9	0.2	45.0	100.0
LIE	24.9	3.7	3.7	18.7	3.5	2.5	5.4	0.0	1.8	0.0	35.6	100.0
DEU	1.0	14.8	31.9	1.1	1.7	3.1	0.6	1.1	1.0	0.9	42.8	100.0
NLD	9.2	3.8	26.9	0.4	1.0	3.6	1.1	0.2	2.2	3.1	48.4	100.0
CHE	6.2	5.0	6.8	0.5	2.9	3.5	0.8	0.4	2.0	0.5	71.4	100.0
JPN	2.4	8.7	45.0	8.9	0.4	1.4	0.4	0.5	0.9	1.9	29.5	100.0
PAN	35.0	9.5	1.2	1.8	1.7	1.1	6.2	0.6	2.0	0.2	40.7	100.0
BRA	12.9	14.1	18.6	1.6	0.6	3.1	0.6	0.6	2.1	2.3	43.4	100.0
Other parents	9.5	19.5	11.7	2.2	3.7	4.0	1.6	1.6	2.1	3.7	40.3	100.0

Row percentages: destination mix of each row group. Foreign MNEs with a recorded parent country; top 10 parents and top 10 destinations by foreign-MNE export value. Pooled 2006–2022, nine LAC origins (Ecuador excluded), all destinations. Parent = country of the multinational's ultimate parent as recorded (Orbis/D&B).

Table 13: Top-10 parents × top-10 destinations, column %

Parent / Destination	CHN	USA	BRA	JPN	ESP	CHL	KOR	CAN	NLD	ARG	Other destinations
USA	14.6	38.7	17.2	18.8	30.1	22.5	17.0	4.1	33.0	48.1	23.0
GBR	43.7	8.1	7.2	29.3	18.8	21.8	43.5	22.8	25.7	1.5	13.7
CAN	6.2	3.8	3.1	9.0	7.8	1.4	5.1	55.1	3.0	0.8	9.1
LIE	8.3	1.5	1.9	21.5	6.1	4.6	10.1	0.1	4.0	0.1	4.9
DEU	0.3	4.7	13.0	1.0	2.4	4.5	0.9	1.7	1.8	1.6	4.7
NLD	2.3	1.1	10.1	0.3	1.3	4.7	1.6	0.2	3.5	5.2	4.9
CHE	1.4	1.3	2.3	0.4	3.3	4.3	1.0	0.6	2.9	0.8	6.6
JPN	0.5	2.2	14.6	6.5	0.5	1.6	0.4	0.6	1.2	2.7	2.6
PAN	7.2	2.4	0.4	1.3	1.8	1.2	7.2	0.8	2.8	0.2	3.5
BRA	2.3	3.0	5.0	1.0	0.6	2.9	0.6	0.6	2.5	2.8	3.2
Other parents	13.4	33.1	25.3	10.9	27.3	30.4	12.6	13.4	19.6	36.0	23.7
Total	100.0	100.0	100.0	100.0	100.0	100.0	100.0	100.0	100.0	100.0	100.0

Column percentages: who supplies each destination. Foreign MNEs with a recorded parent country; top 10 parents and top 10 destinations by foreign-MNE export value. Pooled 2006–2022, nine LAC origins (Ecuador excluded), all destinations. Parent = country of the multinational's ultimate parent as recorded (Orbis/D&B).

Table 14: Foreign-MNE exports: destination region by LAC origin (row %)

Origin	Destination region	Latin America	North America	Europe	Asia	Africa	Oceania	Rest of World	Total
ARG		45.5	9.0	15.5	23.0	6.2	0.8	0.0	292.3
CHL		13.1	11.6	13.3	60.5	0.3	1.2	0.0	229.3
COL		40.7	19.7	25.4	12.6	0.5	0.3	0.9	186.5
CRI		33.0	40.7	16.2	9.8	0.1	0.1	0.0	62.5
DOM		20.3	59.3	9.4	10.5	0.2	0.2	0.1	47.1
PER		15.4	12.9	25.3	44.9	1.0	0.4	0.0	183.3
PRY		79.4	1.3	11.6	6.6	1.1	0.0	0.0	40.0
SLV		46.8	48.1	4.3	0.6	0.1	0.1	0.0	31.0
URY		39.1	7.7	20.5	28.0	3.0	0.1	1.6	37.2

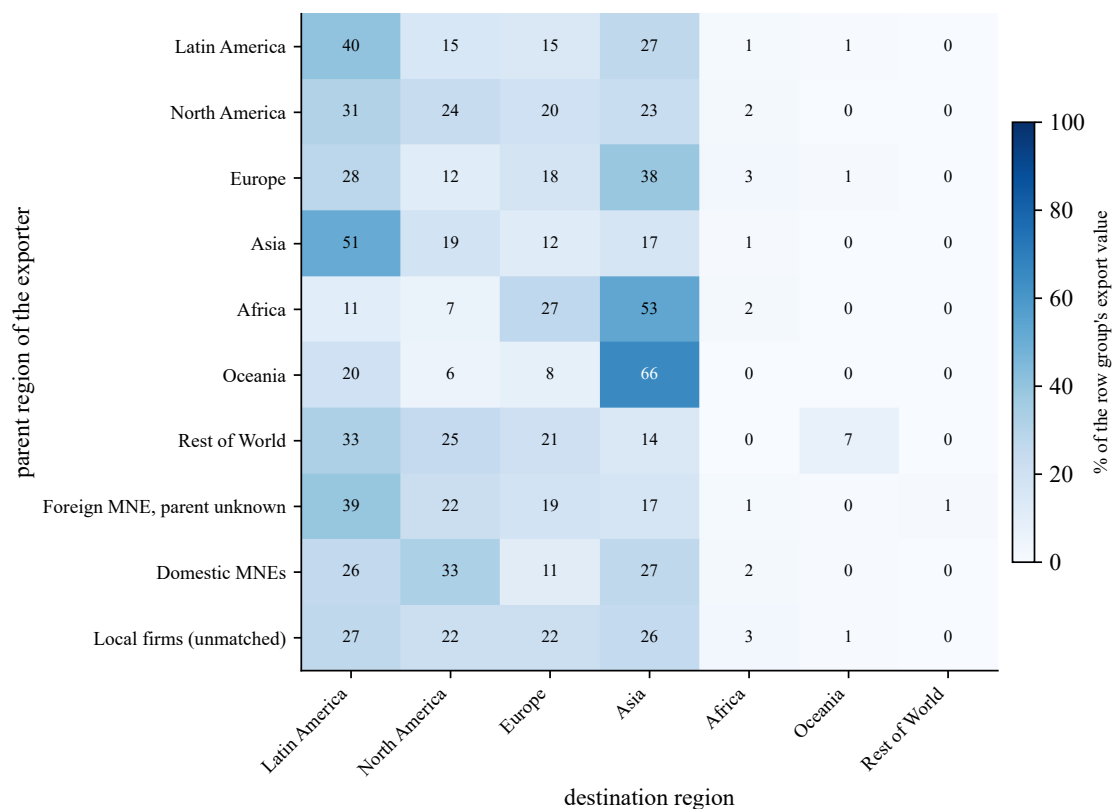
Foreign-MNE exports; row percentages; last column = row total, USD bn.

Table 15: Local (unmatched) firms: destination region by LAC origin (row %)

Origin	Destination region	Latin America	North America	Europe	Asia	Africa	Oceania	Rest of World	Total
ARG		32.2	8.3	21.4	28.5	8.4	1.3	0.0	260.8
CHL		15.5	18.8	20.4	44.3	0.6	0.5	0.0	184.3
COL		32.2	32.8	23.5	8.5	0.8	0.3	1.9	135.0
CRI		32.1	35.4	26.3	5.4	0.3	0.5	0.0	38.4
DOM		31.9	46.8	15.7	4.6	0.6	0.0	0.3	20.9
PER		18.3	28.0	24.6	28.1	0.3	0.4	0.2	221.5
PRY		64.6	3.6	20.0	10.6	1.1	0.0	0.0	24.7
SLV		44.9	46.3	4.6	3.7	0.1	0.3	0.1	34.2
URY		38.7	5.2	21.6	26.0	7.3	0.1	1.3	26.2

Local (unmatched) firms exports; row percentages; last column = row total, USD bn.

4.3 Variant: heat maps (item 1d)

**Figure 15:** Parent region × destination region, row %.

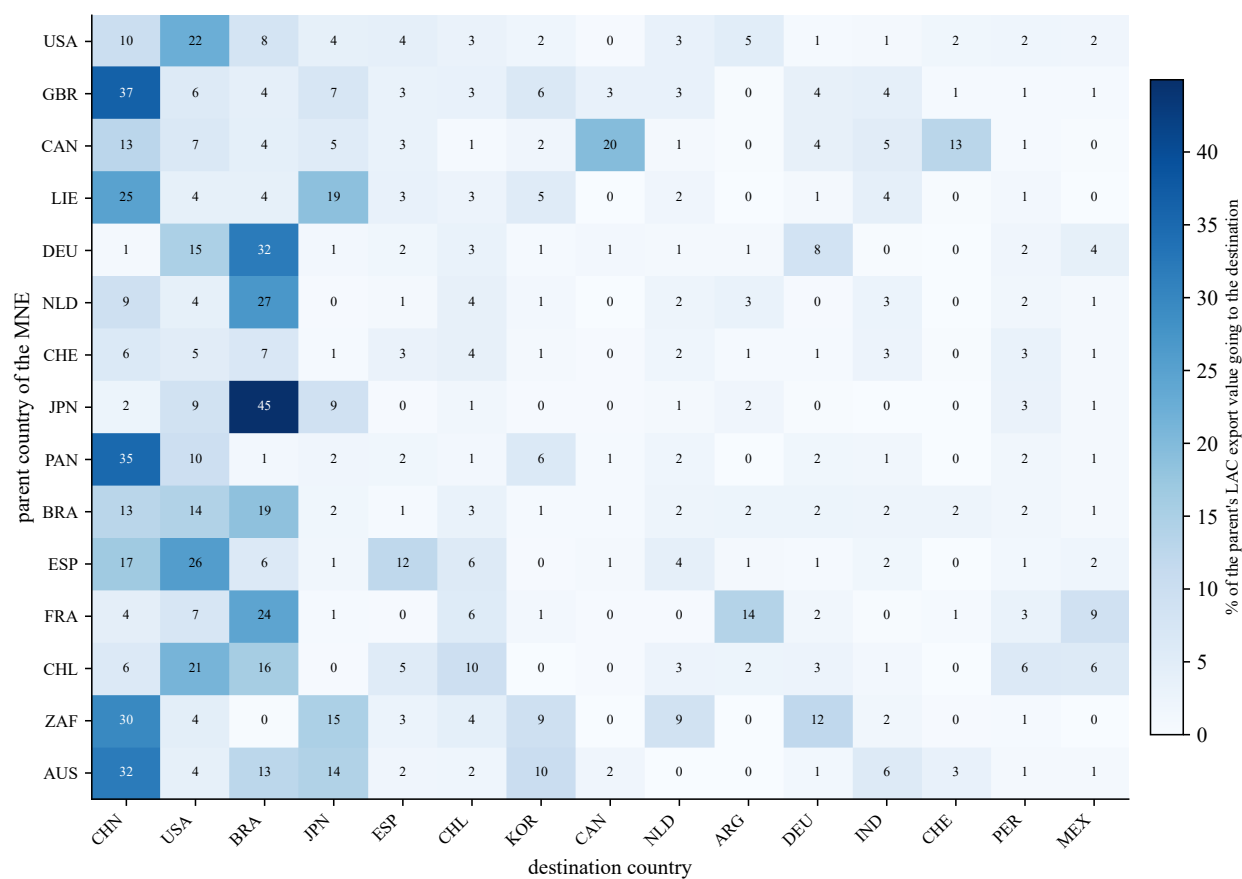


Figure 16: Top-15 parents × top-15 destinations: % of the parent's export value going to each destination.

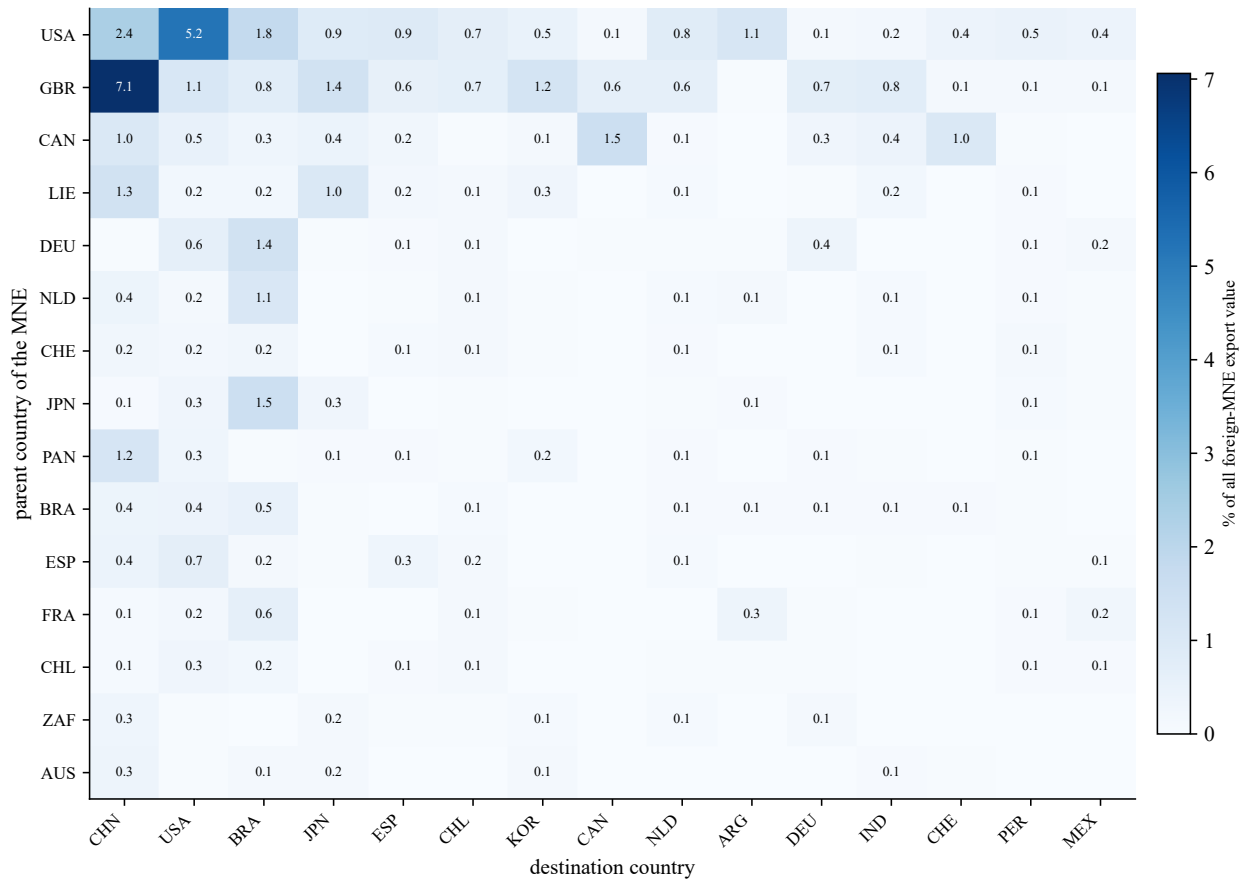


Figure 17: Top-15 parents × top-15 destinations: cell share of all foreign-MNE export value (%).

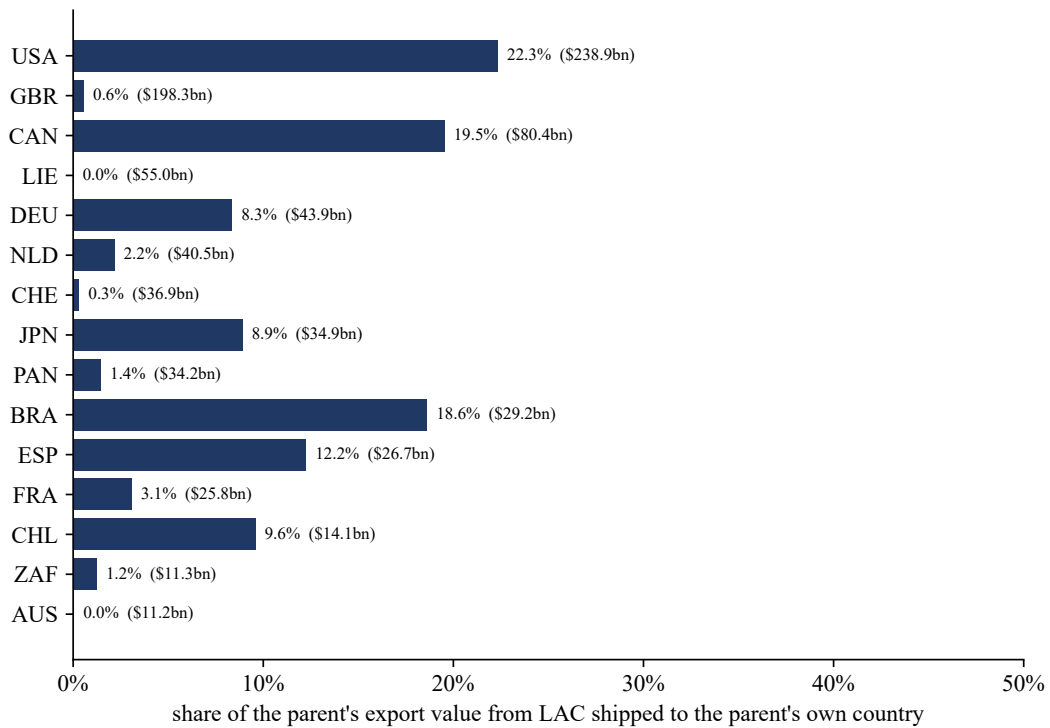


Figure 18: Share of each parent's export value shipped to the parent's own country.

5 Fact 4 — A small set of large multinational groups accounts for the bulk of exports

5.1 Original exhibits

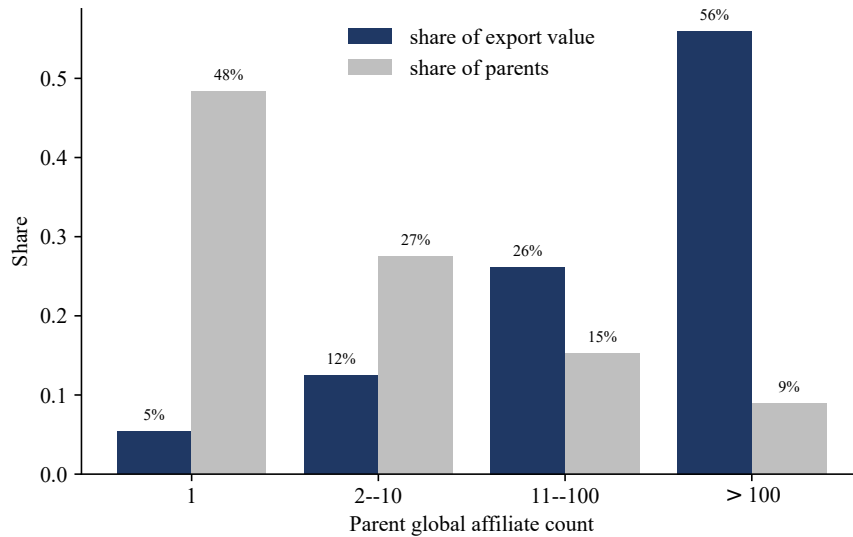


Figure 19: Figure 5 of the note, reproduced: foreign-MNE export value and parent counts by global affiliate-network size.

Table 16: Numbers behind Figure 5

Global affiliate count	Parents	Share of parents (%)	Export value (\$bn)	Share of value (%)
1	5,982	48.4	54.3	5.4
2-10	3,397	27.5	124.4	12.4
11-100	1,893	15.3	261.7	26.2
>100	1,099	8.9	560.2	56.0
All parents with a network record	12,371	100	1,000.6	100

Foreign-MNE parents (parent name key) exporting from the nine origins; 98% of parents and 97% of foreign-MNE export value have a matched global-network record (Orbis $\cup$ D&B roster, src/16). Network size = number of distinct worldwide affiliates.

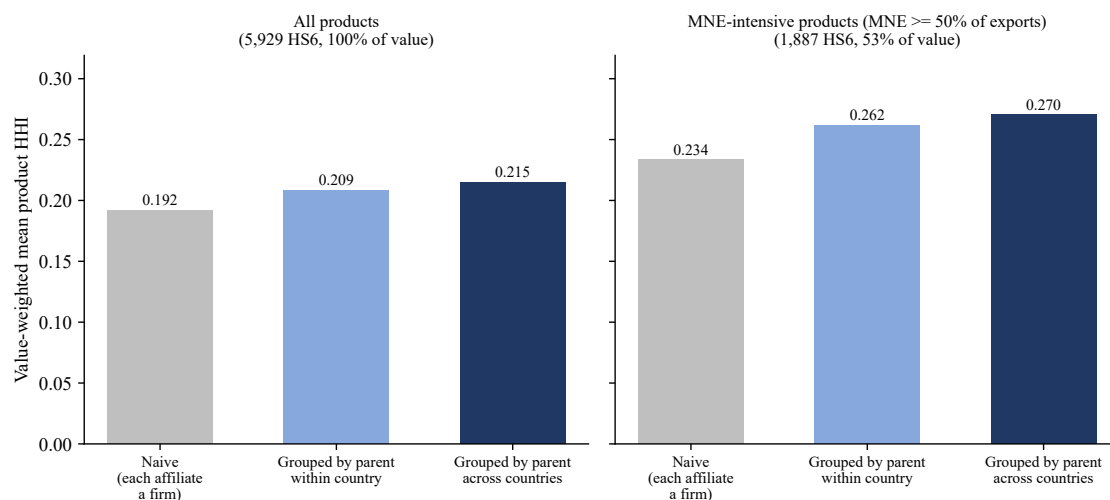


Figure 20: Figure 6 of the note, reproduced: product-level export concentration, naive firm count vs grouping affiliates by parent.

Table 17: Numbers behind Figure 6: HHI, top-exporter share, effective number of exporters

Sample	Measure	Naive	Within country	Across countries
<i>All products: 5,929 HS6, 100% of value</i>				
	Mean HHI	0.192	0.209	0.215
	Mean top-exporter share	31.0%	32.5%	33.7%
	Effective number of exporters (1/HHI)	12.5	11.7	10.4
<i>MNE-intensive products (MNE $\geq$ 50% of exports): 1,887 HS6, 53% of value</i>				
	Mean HHI	0.234	0.262	0.270
	Mean top-exporter share	36.0%	38.7%	39.8%
	Effective number of exporters (1/HHI)	8.9	8.0	7.4

Value-weighted means across HS6 products (nine origins, Ecuador excluded). Naive: each exporting affiliate (Tax ID) is a firm; then affiliates of the same global parent are grouped within a country and across countries.

5.2 Variant: groups versus affiliates inside the market cells (item 1e, “MNE cantidad”)

Table 18: Affiliates per group in the origin–destination–product–year cells

Origin-destination-product-year cells with ≥ 1 MNE	
Cells	1,008,755
MNE affiliates per cell (mean)	1.91
MNE groups per cell (mean)	1.90
Affiliates per group, pooled	1.01
Affiliates per group, value-weighted mean	1.08
Cells where one group exports through > 1 affiliate (%)	1.0

Group = ultimate parent (Orbis GUO id or D&B global ultimate; parent name when the id is missing).

Table 19: Fact-5 regressions with ln(# MNE groups) in place of ln(# MNE firms)

	OxYr + DxYr + P FE			ODP + ODY FE		
	(1)	(2)	(3)	(4)	(5)	(6)
<i>Panel A: all exports (ln)</i>						
ln(# MNE firms)	2.0685*** (0.0368)			0.9414*** (0.0206)		
ln(# MNE groups)		2.0728*** (0.0371)	2.0617*** (0.0365)		0.9416*** (0.0208)	0.9421*** (0.0208)
ln(affiliates per group)			2.7723*** (0.2413)			0.8692*** (0.0648)
<i>Panel B: non-MNE exports (ln)</i>						
ln(# MNE firms)	1.3406*** (0.0319)			0.2389*** (0.0178)		
ln(# MNE groups)		1.3471*** (0.0319)	1.3427*** (0.0317)		0.2400*** (0.0177)	0.2400*** (0.0177)
ln(affiliates per group)			1.1055*** (0.2963)			0.0605 (0.0985)
Product FE	✓	✓	✓			
Origin $\times$ year FE	✓	✓	✓			
Destination $\times$ year FE	✓	✓	✓			
Origin $\times$ dest. $\times$ product FE				✓	✓	✓
Origin $\times$ dest. $\times$ year FE				✓	✓	✓
Observations (all exports)	1,008,185	1,008,185	1,008,185	883,545	883,545	883,545
Observations (non-MNE exports)	475,514	475,514	475,514	429,882	429,882	429,882

Cells with at least one MNE; dep. var. ln exports. ‘MNE firms’ counts matched exporting affiliates; ‘MNE groups’ counts distinct ultimate parents. SE clustered at origin-destination. *** p<0.01, ** p<0.05, * p<0.1

6 Fact 5 — Greater multinational presence is associated with higher trade volumes

6.1 Original exhibit

Table 20: Table 1 of the note, reproduced: multinational presence and trade volume (intensive and extensive margins)

	Intensive: ln(# MNE firms)			Extensive: any MNE present		
	(1)	(2)	(3)	(1)	(2)	(3)
<i>Panel A: all exports (ln)</i>						
ln(# MNE firms)	2.0823*** (0.0363)	2.0685*** (0.0368)	0.9414*** (0.0206)			
Any MNE present				1.4622*** (0.0394)	1.4591*** (0.0388)	0.6818*** (0.0165)
<i>Panel B: non-MNE exports (ln)</i>						
ln(# MNE firms)	1.3536*** (0.0318)	1.3406*** (0.0319)	0.2389*** (0.0178)			
Any MNE present				1.4168*** (0.0371)	1.3996*** (0.0367)	0.1311*** (0.0102)
Origin FE	✓			✓		
Destination FE	✓			✓		
Product FE	✓	✓		✓	✓	
Year FE	✓			✓		
Origin × year FE		✓			✓	
Destination × year FE		✓			✓	
Origin × dest. × product FE			✓			✓
Origin × dest. × year FE			✓			✓
Observations (all exports)	1,008,472	1,008,185	883,545	1,829,852	1,829,626	1,633,319
Observations (non-MNE exports)	475,856	475,514	429,882	1,297,322	1,297,111	1,137,550

Origin-destination-product-year cells; dep. var. ln exports. Intensive margin: ln number of MNE firms (cells with ≥ 1 MNE); extensive margin: indicator for any MNE present. Reproduction of the note's Table 1 on the current base. SE clustered at origin-destination. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$

6.2 Variant: foreign vs domestic MNEs; foreign MNEs present through headquarters vs not (item 1e)

Table 21: MNE export value by origin: domestic share, and foreign-MNE value by the group's presence at the destination

Origin	MNE value (\$bn)	of which domestic (%)	Foreign-MNE value (\$bn)	% of foreign-MNE value: group present at destination	
				through headquarters	not through HQ (of which via another affiliate)
ARG	337.1	13.3	292.3	6.5	93.5 (41.8)
CHL	294.4	22.1	229.3	4.3	95.7 (56.7)
COL	393.0	52.5	186.5	6.5	93.5 (34.2)
CRI	66.3	5.7	62.5	25.5	74.5 (33.8)
DOM	48.0	1.8	47.1	29.0	71.0 (21.6)
PER	193.0	5.0	183.3	3.9	96.1 (28.8)
PRY	40.7	1.8	40.0	2.1	97.9 (65.5)
SLV	31.2	0.6	31.0	40.7	59.3 (28.1)
URY	44.2	15.9	37.2	5.2	94.8 (37.7)
All	1,447.8	23.4	1,109.2	8.4	91.6 (40.5)

'Through headquarters': the affiliate ships to its parent's country. 'Not through HQ': every other destination; in parentheses the part where the group has another affiliate in the destination (Orbis/D&B roster). Pooled 2006–2022.

Table 22: Intensive margin: $\ln(\# \text{ MNE firms})$ decomposed into foreign / domestic and foreign through HQ / not through HQ

	All MNEs		Foreign / domestic		Foreign through HQ / not through HQ / domestic	
	(1)	(2)	(3)	(4)	(5)	(6)
<i>Panel A: all exports (ln)</i>						
$\ln(\# \text{ MNE firms})$	2.0685*** (0.0368)	0.9414*** (0.0206)				
$\ln(\# \text{ foreign MNEs})$			1.2129*** (0.0285)	0.4506*** (0.0124)		
$\ln(\# \text{ foreign MNEs present through HQ})$					0.3970*** (0.0856)	0.1448*** (0.0370)
$\ln(\# \text{ foreign MNEs not present through HQ})$					0.9465*** (0.0536)	0.3133*** (0.0382)
$\ln(\# \text{ domestic MNEs})$			0.4382*** (0.0259)	0.2214*** (0.0151)	0.3770*** (0.0514)	0.1555*** (0.0315)
<i>Panel B: non-MNE exports (ln)</i>						
$\ln(\# \text{ MNE firms})$	1.3406*** (0.0319)	0.2389*** (0.0178)				
$\ln(\# \text{ foreign MNEs})$			1.1331*** (0.0375)	0.2413*** (0.0217)		
$\ln(\# \text{ foreign MNEs present through HQ})$					0.2934*** (0.0882)	0.1945*** (0.0527)
$\ln(\# \text{ foreign MNEs not present through HQ})$					1.0330*** (0.0607)	0.1951*** (0.0648)
$\ln(\# \text{ domestic MNEs})$			0.4707*** (0.0341)	0.1250*** (0.0218)	0.4142*** (0.0633)	0.0946* (0.0552)
Product FE	✓		✓		✓	
Origin $\times$ year FE	✓		✓		✓	
Destination $\times$ year FE	✓		✓		✓	
Origin $\times$ dest. $\times$ product FE		✓		✓		✓
Origin $\times$ dest. $\times$ year FE		✓		✓		✓
Observations (all exports)	1,008,185	883,545	113,422	98,223	9,197	7,524
Observations (non-MNE exports)	475,514	429,882	93,010	81,563	8,780	7,255

Origin-destination-product-year cells; dep. var. $\ln$ exports. Intensive margin: plain logs of the counts, so each column uses the cells where every count entering it is positive (no $\ln(1+x)$). (1)–(2) all MNEs; (3)–(4) foreign vs domestic; (5)–(6) foreign split by whether the group is present at the destination through its headquarters (the parent's country). SE clustered at origin-destination. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$

Table 23: Extensive margin: presence dummies decomposed into foreign / domestic and foreign through HQ / not through HQ

	All MNEs		Foreign / domestic		Foreign through HQ / not through HQ / domestic	
	(1)	(2)	(3)	(4)	(5)	(6)
<i>Panel A: all exports (ln)</i>						
Any MNE present	1.4591*** (0.0388)	0.6818*** (0.0165)				
Any foreign MNE			1.4990*** (0.0339)	0.6986*** (0.0158)		
Any foreign MNE present through HQ					1.3873*** (0.0951)	0.5909*** (0.0431)
Any foreign MNE not present through HQ					1.5093*** (0.0319)	0.6976*** (0.0146)
Any domestic MNE			1.4670*** (0.0416)	0.4917*** (0.0165)	1.4334*** (0.0388)	0.4909*** (0.0166)
<i>Panel B: non-MNE exports (ln)</i>						
Any MNE present	1.3996*** (0.0367)	0.1311*** (0.0102)				
Any foreign MNE			1.2983*** (0.0346)	0.1356*** (0.0106)		
Any foreign MNE present through HQ					1.0183*** (0.0900)	0.1265*** (0.0194)
Any foreign MNE not present through HQ					1.2782*** (0.0326)	0.1386*** (0.0107)
Any domestic MNE			1.1316*** (0.0428)	0.1370*** (0.0138)	1.0959*** (0.0403)	0.1367*** (0.0138)
Product FE	✓		✓		✓	
Origin × year FE	✓		✓		✓	
Destination × year FE	✓		✓		✓	
Origin × dest. × product FE		✓		✓		✓
Origin × dest. × year FE		✓		✓		✓
Observations (all exports)	1,829,626	1,633,319	1,829,626	1,633,319	1,829,626	1,633,319
Observations (non-MNE exports)	1,297,111	1,137,550	1,297,111	1,137,550	1,297,111	1,137,550

Origin-destination-product-year cells; dep. var. ln exports. Extensive margin: indicators for the presence of at least one MNE of each type in the cell; all cells. SE clustered at origin-destination. *** p<0.01, ** p<0.05, * p<0.1

7 Fact 6 — Distance is a weaker barrier to trade for multinational corporations

7.1 Original exhibit

Table 24: Table 2 of the note, reproduced: distance and firm exports, multinationals split by presence at the destination

Dep. var.: ln exports	(1)	(2)	(3)	(4)
ln distance	-0.1639*** (0.0454)	-0.1848*** (0.0460)	-0.1632*** (0.0452)	-0.1838*** (0.0459)
ln distance × MNE	0.0461*** (0.0038)	0.0458*** (0.0038)		
ln distance × MNE, present			0.0689*** (0.0070)	0.0682*** (0.0070)
ln distance × MNE, not present			0.0393*** (0.0032)	0.0391*** (0.0032)
Origin FE	✓		✓	
Destination FE	✓		✓	
Year FE	✓		✓	
Origin × year FE		✓		✓
Destination × year FE		✓		✓
Product FE	✓	✓	✓	✓
Observations	5,081,547	5,081,360	5,081,547	5,081,360

Distance and firm exports. Firm×origin×destination×product×year; non-MNE flows the omitted base. Reproduction of the note's Table 2 on the current base: multinationals (foreign or domestic) split by presence at the destination (the group has an affiliate there, or the destination is the parent's country). SE clustered at origin-destination. *** p<0.01, ** p<0.05, * p<0.1

7.2 Variant: foreign vs domestic MNEs; foreign MNEs present through headquarters vs not (item 1e)

Table 25: Distance and firm exports: foreign / domestic, then foreign through HQ / not through HQ

Dep. var.: ln exports	(1)	(2)	(3)	(4)
ln distance	-0.1645*** (0.0454)	-0.1853*** (0.0460)	-0.1644*** (0.0454)	-0.1851*** (0.0460)
ln distance × foreign MNE	0.0487*** (0.0041)	0.0483*** (0.0040)		
ln distance × foreign MNE, present through HQ			0.0642*** (0.0159)	0.0638*** (0.0161)
ln distance × foreign MNE, not present through HQ			0.0477*** (0.0035)	0.0473*** (0.0035)
ln distance × domestic MNE	0.0318*** (0.0042)	0.0318*** (0.0042)	0.0317*** (0.0042)	0.0317*** (0.0042)
Origin FE	✓		✓	
Destination FE	✓		✓	
Year FE	✓		✓	
Origin × year FE		✓		✓
Destination × year FE		✓		✓
Product FE	✓	✓	✓	✓
Observations	5,081,547	5,081,360	5,081,547	5,081,360

Distance and firm exports. Firm×origin×destination×product×year; non-MNE flows the omitted base. (1)–(2) foreign vs domestic multinationals; (3)–(4) foreign MNEs split by whether the group is present at the destination through its headquarters (the parent's country) or not. SE clustered at origin-destination. *** p<0.01, ** p<0.05, * p<0.1

8 Products at HS6, with descriptions (item 1f)

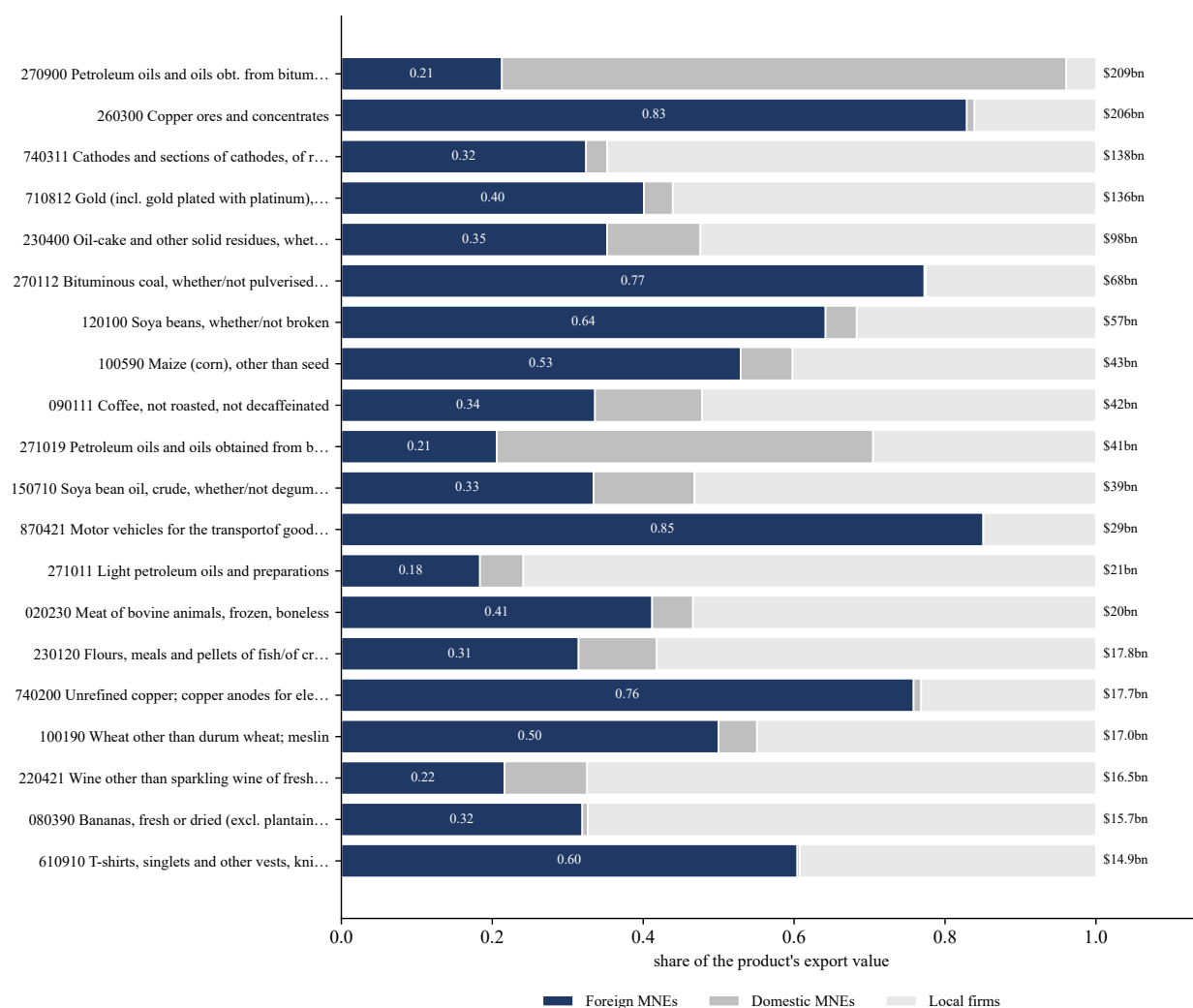


Figure 21: Top-20 HS6 products by export value: foreign MNE / domestic MNE / local shares.

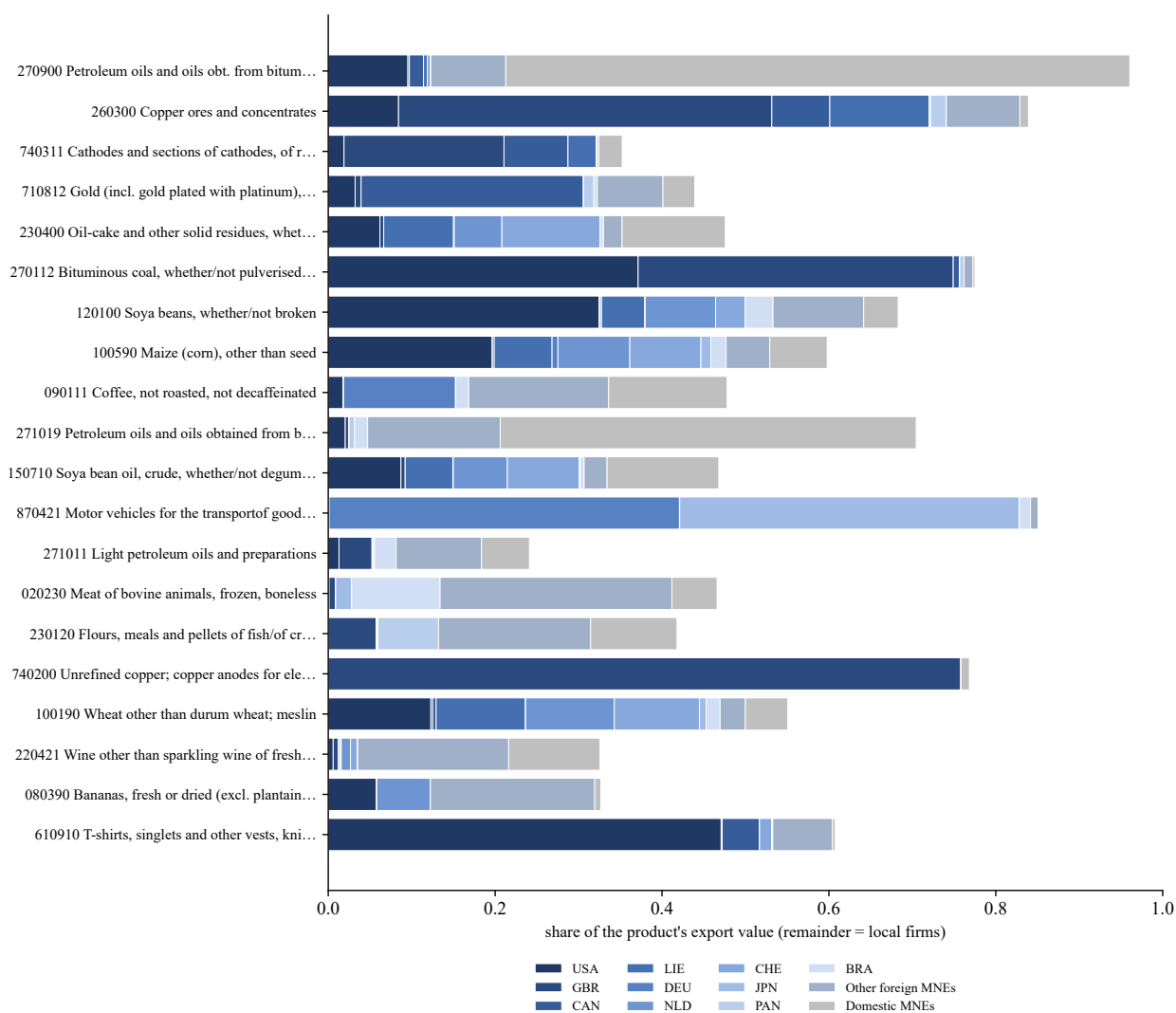


Figure 22: Top-20 HS6 products: foreign bar split by parent country (remainder = local firms).

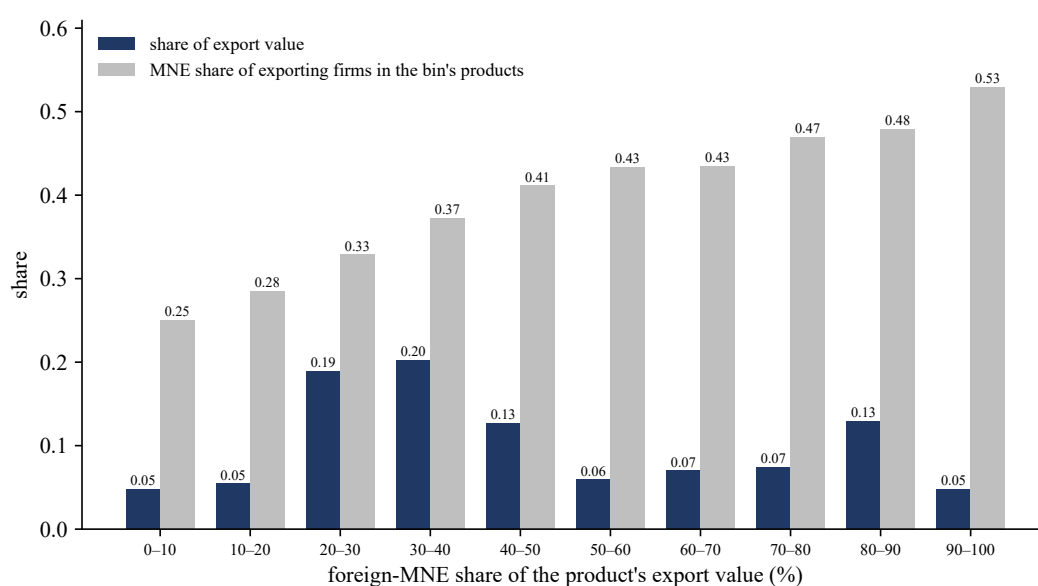


Figure 23: Distribution of the HS6 foreign-MNE share: share of export value, and MNE share of exporting firms, by bin.

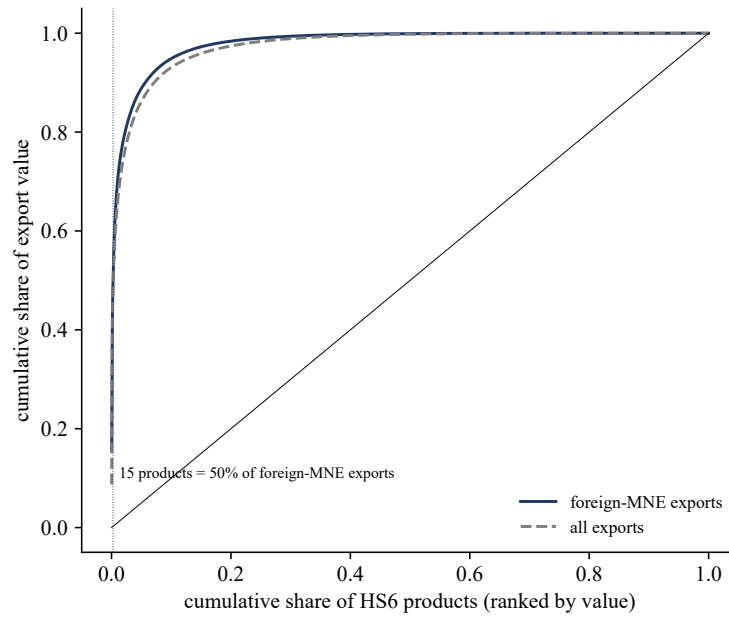


Figure 24: Concentration of exports across HS6 products: foreign-MNE exports vs all exports.

Table 26: HS sections by export value: MNE shares and leading parent

HS section	\$bn	For.	Dom.	Local	Lead parent
V Mineral products	655.5	51	29	19	GBR (0.41)
II Vegetable products	338.5	42	7	51	USA (0.28)
IV Prepared foodstuffs, beverages, tobacco	253.8	37	10	53	USA (0.19)
XV Base metals	238.4	41	3	56	GBR (0.41)
XIV Precious metals and stones	154.7	41	4	56	CAN (0.62)
I Live animals; animal products	131.1	34	14	51	SAU (0.16)
VI Chemicals	121.8	49	14	37	USA (0.22)
XVII Transport equipment	90.2	78	4	18	DEU (0.30)
XI Textiles and apparel	74.9	49	3	48	USA (0.52)
III Fats and oils	65.0	31	10	59	CHE (0.24)
XVI Machinery and electrical equipment	62.7	63	4	33	USA (0.39)
VII Plastics and rubber	61.7	51	8	42	USA (0.31)
X Pulp, paper	41.2	54	28	18	PAN (0.53)
XVIII Optical, precision instruments	32.7	72	8	19	USA (0.61)
IX Wood and articles	23.6	27	34	39	BHS (0.21)
VIII Hides, skins, leather	15.3	40	12	49	BGR (0.15)
XIII Stone, ceramics, glass	10.2	49	4	46	USA (0.26)
XX Miscellaneous manufactures	10.1	53	7	40	USA (0.41)
Unclassified	6.7	49	29	23	USA (0.31)
XII Footwear, headgear	4.6	48	1	50	USA (0.58)
XXI Works of art	0.6	58	1	41	BGR (0.50)
XIX Arms	0.3	1	0	99	CHE (0.51)
Special / unclassified (98-99)	0.1	13	1	87	USA (0.45)

HS sections ranked by export value. For./Dom./Local = foreign-MNE, domestic-MNE and local-firm shares of the section's export value, percent; Lead parent = largest parent country among the section's foreign MNEs and its share of the section's foreign-MNE value.

Table 27: Top 30 HS6 products by export value

HS6	Description	\$bn	For.	Dom.	Local	Lead parent
270900	Petroleum oils and oils obt. from bituminous mins., crude	209.3	21	75	4	USA (0.45)
260300	Copper ores and concentrates	206.1	83	1	16	GBR (0.54)
740311	Cathodes and sections of cathodes, of refined copper, unwrought	137.9	32	3	65	GBR (0.59)
710812	Gold (incl. gold plated with platinum), in unwrought forms (excl. powder)	135.7	40	4	56	CAN (0.66)
230400	Oil-cake and other solid residues, whether/not ground/in pellets, from extraction of soyabean ...	97.5	35	12	52	CHE (0.33)
270112	Bituminous coal, whether/not pulverised but not agglomerated	68.1	77	0	23	GBR (0.49)
120100	Soya beans, whether/not broken	56.6	64	4	32	USA (0.51)
100590	Maize (corn), other than seed	43.1	53	7	40	USA (0.37)
090111	Coffee, not roasted, not decaffeinated	42.4	34	14	52	DEU (0.40)
271019	Petroleum oils and oils obtained from bituminous minerals (other than crude) and preparations ...	40.9	21	50	30	ESP (0.30)
150710	Soya bean oil, crude, whether/not degummed, not chemically modified	38.7	33	13	53	USA (0.26)
870421	Motor vehicles for the transport of goods (excl. of 8704.10), with C-I internal combustion pist...	29.1	85	0	15	DEU (0.49)
271011	Light petroleum oils and preparations	21.4	18	6	76	GBR (0.22)
020230	Meat of bovine animals, frozen, boneless	20.2	41	5	53	SAU (0.48)
230120	Flours, meals and pellets of fish/of crustaceans, molluscs/other aquatic invertebrates	17.8	31	10	58	BMU (0.34)
740200	Unrefined copper; copper anodes for electrolytic refining	17.7	76	1	23	GBR (1.00)
100190	Wheat other than durum wheat; meslin	17.0	50	5	45	USA (0.25)
220421	Wine other than sparkling wine of fresh grapes, incl. fortified; grape must with fermentation ...	16.5	22	11	67	FRA (0.15)
080390	Bananas, fresh or dried (excl. plantains) [HS2012 code]	15.7	32	1	67	JOR (0.34)
610910	T-shirts, singlets and other vests, knitted/crocheted, of cotton	14.9	60	0	39	USA (0.78)
260800	Zinc ores and concentrates	14.7	72	0	27	GBR (0.67)
080610	Grapes, fresh	13.3	29	9	62	ESP (0.19)
260111	Iron ores and concentrates (excl. roasted iron pyrites), non-agglomerated	13.1	2	47	51	CAN (0.64)
260700	Lead ores and concentrates	12.9	58	0	42	GBR (0.69)
870323	Vehicles (excl. of 87.02 and 8703.10) principally designed for the transport of persons, with s...	12.9	75	0	25	FRA (0.45)
020130	Meat of bovine animals, fresh/chilled, boneless	12.7	35	11	55	SAU (0.52)
901890	Instruments and appliances used in medical/surgical/veterinary sciences, incl. other electro-m...	12.2	89	0	11	USA (0.48)
300490	Medicaments (excluding goods of heading 30.02/30.05/30.06/3004.10-3004.50) consisting of mixed...	10.9	55	7	38	USA (0.24)
382490	Chemical products and preparations of the chemical/allied industries (including those consisting...	10.8	36	14	50	LIE (0.60)
470321	Chemical wood pulp, soda/sulphate, other than dissolving grades, semi-bleached/bleached, confi...	9.4	71	29	0	PAN (0.82)

Top 30 HS6 products by export value (57% of the scope's exports). For./Dom./Local = foreign-MNE, domestic-MNE and local-firm shares of the product's export value, percent (pooled 2006–2022, nine LAC origins). Foreign = matched firms whose parent is abroad or unknown; Domestic = parent in the exporting country; Local = unmatched. Lead parent = largest parent country among the product's foreign MNEs and its share of the product's foreign-MNE value.

Table 28: Top 30 HS6 products by foreign-MNE export value

HS6	Description	\$bn	For.	Dom.	Local	Lead parent
260300	Copper ores and concentrates	206.1	83	1	16	GBR (0.54)
710812	Gold (incl. gold plated with platinum), in unwrought forms (excl. powder)	135.7	40	4	56	CAN (0.66)
270112	Bituminous coal, whether/not pulverised but not agglomerated	68.1	77	0	23	GBR (0.49)
740311	Cathodes and sections of cathodes, of refined copper, unwrought	137.9	32	3	65	GBR (0.59)
270900	Petroleum oils and oils obt. from bituminous mins., crude	209.3	21	75	4	USA (0.45)
120100	Soya beans, whether/not broken	56.6	64	4	32	USA (0.51)
230400	Oil-cake and other solid residues, whether/not ground/in pellets, from extraction of soyabean ...	97.5	35	12	52	CHE (0.33)
870421	Motor vehicles for the transport of goods (excl. of 8704.10), with C-I internal combustion pist...	29.1	85	0	15	DEU (0.49)
100590	Maize (corn), other than seed	43.1	53	7	40	USA (0.37)
090111	Coffee, not roasted, not decaffeinated	42.4	34	14	52	DEU (0.40)
740200	Unrefined copper; copper anodes for electrolytic refining	17.7	76	1	23	GBR (1.00)
150710	Soya bean oil, crude, whether/not degummed, not chemically modified	38.7	33	13	53	USA (0.26)
901890	Instruments and appliances used in medical/surgical/veterinary sciences, incl. other electrom...	12.2	89	0	11	USA (0.48)
260800	Zinc ores and concentrates	14.7	72	0	27	GBR (0.67)
870323	Vehicles (excl. of 87.02 and 8703.10) principally designed for the transport of persons, with s...	12.9	75	0	25	FRA (0.45)
610910	T-shirts, singlets and other vests, knitted/crocheted, of cotton	14.9	60	0	39	USA (0.78)
271111	Natural gas, liquefied	8.5	100	0	0	USA (1.00)
100190	Wheat other than durum wheat; meslin	17.0	50	5	45	USA (0.25)
720260	Ferro-nickel, in granular/powder form	8.5	100	0	0	ZAF (0.85)
271019	Petroleum oils and oils obtained from bituminous minerals (other than crude) and preparations ...	40.9	21	50	30	ESP (0.30)
020230	Meat of bovine animals, frozen, boneless	20.2	41	5	53	SAU (0.48)
901839	Catheters, cannulae and the like	9.2	89	0	11	USA (0.86)
854231	Electronic integrated circuits, processors and controllers, whether/not combined with memorie...	7.6	99	0	1	USA (0.98)
260700	Lead ores and concentrates	12.9	58	0	42	GBR (0.69)
870322	Vehicles (excl. of 87.02 and 8703.10) principally designed for the transport of persons, with s...	7.5	100	0	0	USA (0.59)
470321	Chemical wood pulp, soda/sulphate, other than dissolving grades, semi-bleached/bleached, conif...	9.4	71	29	0	PAN (0.82)
210690	Food preparations, n.e.s.	7.8	81	1	18	USA (0.77)
300490	Medicaments (excluding goods of heading 30.02/30.05/30.06/3004.10-3004.50) consisting of mixed...	10.9	55	7	38	USA (0.24)
870840	Gear boxes and parts thereof, of the motor vehicles of headings 87.01 to 87.05.	6.0	99	0	1	DEU (0.74)
261610	Silver ores and concentrates	7.6	76	0	24	GBR (0.39)

Top 30 HS6 products by foreign-MNE export value (60% of foreign-MNE exports). For./Dom./Local = foreign-MNE, domestic-MNE and local-firm shares of the product's export value, percent (pooled 2006–2022, nine LAC origins). Foreign = matched firms whose parent is abroad or unknown; Domestic = parent in the exporting country; Local = unmatched. Lead parent = largest parent country among the product's foreign MNEs and its share of the product's foreign-MNE value.

Table 29: HS6 products ($\geq$ \$500m) with the highest foreign-MNE share

HS6	Description	\$bn	For.	Dom.	Local	Lead parent
720260	Ferro-nickel, in granular/powder form	8.46	100	0	0	ZAF (0.85)
790500	Zinc plates, sheets, strip and foil	0.51	100	0	0	GBR (0.01)
790112	Zinc, not alloyed, unwrought, containing by weight <99.99% of zinc	2.23	100	0	0	BRA (1.00)
870321	Vehicles (excl. of 87.02 and 8703.10) principally designed for the transport of persons, with s...	4.78	100	0	0	NLD (0.49)
390410	Poly(vinyl chloride), not mixed with any other substance, in primary forms	4.37	100	0	0	MEX (0.83)
271111	Natural gas, liquefied	8.50	100	0	0	USA (1.00)
870322	Vehicles (excl. of 87.02 and 8703.10) principally designed for the transport of persons, with s...	7.47	100	0	0	USA (0.59)
870333	Vehicles principally designed for the transport of persons (excl. of 87.02 and 8703.10-8703.24)...	3.49	100	0	0	JPN (1.00)
251010	Natural calcium phosphates, natural aluminium calcium phosphates and phosphatic chalk, unground	2.78	100	0	0	BRA (1.00)
280300	Carbon (carbon blacks and other forms of carbon, n.e.s.)	0.81	100	0	0	USA (0.99)
470311	Chemical wood pulp, soda/sulphate, other than dissolving grades, unbleached, coniferous	1.90	100	0	0	PAN (0.99)
300691	Appliances identifiable for ostomy use	1.55	100	0	0	GBR (0.80)
853620	Automatic circuit breakers, for a voltage not >1000V	3.61	99	0	1	IRL (0.88)
854231	Electronic integrated circuits, processors and controllers, whether/not combined with memorie...	7.62	99	0	1	USA (0.98)
381512	Supported catalysts, with precious metal/precious metal comps. as the active substance	0.73	99	0	1	GBR (0.80)
870840	Gear boxes and parts thereof, of the motor vehicles of headings 87.01 to 87.05.	5.98	99	0	1	DEU (0.74)
620711	Men's/boys' underpants and briefs (excl. knitted/crocheted), of cotton	0.53	98	0	2	USA (0.93)
854430	Ignition wiring sets and other wiring sets of a kind used in vehicles/aircraft/ships	2.24	98	0	2	JPN (0.81)
740710	Bars, rods and profiles, of refined copper	0.63	98	0	1	MEX (0.06)
390120	Polyethylene having a sp.gr. of 0.94/more, in primary forms	1.16	98	0	2	USA (0.98)
961900	Sanitary towels and tampons, napkins and napkin liners for babies and similar sanitary article...	2.39	97	1	1	USA (0.79)
350300	Gelatin, incl. gelatin in rectangular (incl. square) sheets, whether/not surface-worked /colou...	0.78	97	0	3	BEL (0.43)
790111	Zinc, not alloyed, unwrought, containing by weight 99.99%/more of zinc	3.48	97	0	3	BRA (0.89)
521213	Woven fabrics of cotton (excl. of 52.08-52.11), dyed, weighing not >200 g/m2	1.04	96	0	4	USA (1.00)
390190	Polymers of ethylene, in primary forms (excl. of 3901.10-3901.30)	1.19	96	0	4	USA (0.99)
840820	Compression-ignition internal combustion piston engines (diesel/semi-diesel engines) of a kind...	1.76	96	0	4	USA (0.54)
761010	Doors, windows and their frames and thresholds for doors, of aluminium	1.53	96	0	4	CYM (0.98)
870422	Motor vehicles for the transport of goods (excl. of 8704.10), with C-I internal combustion pist...	0.95	96	0	4	JPN (0.87)
281700	Zinc oxide; zinc peroxide	0.74	95	0	5	BRA (0.01)
300610	Sterile surgical catgut, similar sterile suture materials (including sterile absorbable surgic...	1.12	95	3	2	IRL (0.95)

HS6 products with export value $\geq$ \$500m ranked by the foreign-MNE share (461 products qualify). For./Dom./Local = foreign-MNE, domestic-MNE and local-firm shares of the product's export value, percent (pooled 2006–2022, nine LAC origins). Foreign = matched firms whose parent is abroad or unknown; Domestic = parent in the exporting country; Local = unmatched. Lead parent = largest parent country among the product's foreign MNEs and its share of the product's foreign-MNE value.

Table 30: HS6 products ($\geq$ \$500m) with the highest domestic-MNE share

HS6	Description	\$bn	For.	Dom.	Local	Lead parent
282520	Lithium oxide and hydroxide	0.5	0	100	0	–
481092	Paper and paperboard(excl. of 4810.13-4810.39), coated on one/both sides with kaolin (China cl...	2.2	1	99	0	IRL (0.75)
310420	Potassium chloride	2.0	3	95	2	DZA (0.55)
441193	Fibreboard of wood/other ligneous materials, whether/not bonded with resins/other organic subs...	0.5	1	94	5	ESP (0.94)
260112	Iron ores and concentrates (excl. roasted iron pyrites), agglomerated	1.8	7	88	4	CHE (1.00)
350110	Casein	0.5	0	84	16	CHL (0.59)
020727	Cuts and edible offal of turkey, frozen	0.5	1	82	17	JPN (0.02)
730640	Tubes, pipes and hollow profiles (excl. of 7306.10-7306.30), welded, of circular cross-section...	0.5	15	82	3	DEU (0.59)
270900	Petroleum oils and oils obt. from bituminous mins., crude	209.3	21	75	4	USA (0.45)
390390	Polymers of styrene, in primary forms (excl. of 3903.11-3903.30)	0.5	15	73	12	NLD (0.59)
440711	Wood sawn/chipped length wise, sliced/peeled, whether/not planed, sanded/end-jointed, of a thi...	2.5	10	71	19	BHS (0.39)
441239	Plywood, consisting solely of sheets of wood (other than bamboo,excl. of 4412.31 and4412.32), ...	2.8	28	71	1	VGB (0.45)
390319	Polystyrene other than expansible, in primary forms	0.6	22	67	11	PAN (0.99)
040610	Fresh (unripened/uncured) cheese, incl. whey cheese, and curd	1.1	12	66	22	MUS (0.34)
283691	Lithium carbonates	7.1	14	65	21	USA (0.54)
540219	High tenacity yarn other than textured yarn/sewing thread, of nylon/other polyamides (excl. of...	0.5	36	64	0	GBR (0.01)
481141	Gummed/adhesive paper,paperboard,coated, impregnated, covered, surface-coloured, surface-decor...	0.8	26	62	12	GBR (0.54)
902139	Artificial parts of the body other than teeth, dental fittings and joints	3.8	10	60	30	VGB (0.45)
722830	Bars and rods of other alloy steel (excl. of 72.27, 7228.10 and 7228.20), not further worked t...	0.7	25	56	19	SGP (0.59)
020329	Meat of swine (excl. carcasses/half-carcasses/hams/shoulders and cuts thereof), frozen	2.6	29	54	17	JPN (0.55)
470329	Chemical wood pulp, soda/sulphate, other than dissolving grades, semi-bleached/bleached, non-c...	7.8	46	54	0	PAN (1.00)
491110	Trade advertising material, commercial catalogue-sandthe like	0.6	16	53	31	USA (0.25)
880240	Aeroplanes and other aircraft, of an unladen weight >15000kg	5.4	40	53	7	CHL (0.99)
410449	Tanned/crust hides and skins of bovine (incl. buffalo)/equine animals, without hair on, in the...	0.7	37	52	12	URY (0.67)
271019	Petroleum oils and oils obtained from bituminous minerals (other than crude) and preparations ...	40.9	21	50	30	ESP (0.30)
210111	Extracts, essences and concentrates of coffee	2.9	23	48	29	CHE (0.28)
260111	Iron ores and concentrates (excl. roasted iron pyrites), non-agglomerated	13.1	2	47	51	CAN (0.64)
330300	Perfumes and toilet waters	1.5	34	46	20	PAN (0.36)
030319	Pacific salmon (Oncorhynchus gorbuscha/keta/tshawytscha/kisutch/masou/rhodurus), frozen (excl...	0.7	8	44	47	CAN (0.39)
391990	Self-adhesive plates, sheets, film, foil, tape, strip and other flat shapes, of plastics, whet...	0.8	41	44	15	GBR (0.50)

HS6 products with export value $\geq$ \$500m ranked by the domestic-MNE share. For./Dom./Local = foreign-MNE, domestic-MNE and local-firm shares of the product's export value, percent (pooled 2006–2022, nine LAC origins). Foreign = matched firms whose parent is abroad or unknown; Domestic = parent in the exporting country; Local = unmatched. Lead parent = largest parent country among the product's foreign MNEs and its share of the product's foreign-MNE value.

Table 31: Distribution of the HS6 foreign share, numbers

Foreign share bin (%)	Share of value	MNE share of exporting firms	N HS6
0–10	0.048	0.250	1,426
10–20	0.054	0.285	731
20–30	0.189	0.329	652
30–40	0.202	0.372	591
40–50	0.127	0.411	519
50–60	0.059	0.433	442
60–70	0.070	0.435	386
70–80	0.074	0.470	311
80–90	0.129	0.479	311
90–100	0.048	0.530	560

MNE share of exporting firms = matched firm-cells (firm x destination x HS6 x year) over all firm-cells in the bin's products.

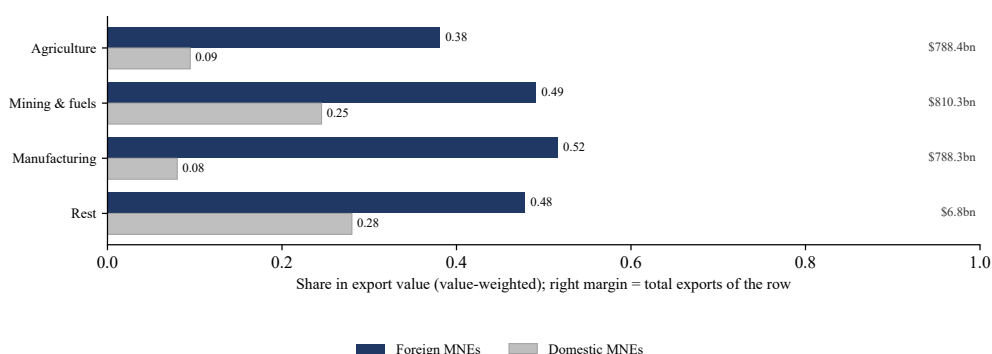
9 The four sectors at a glance (item 2)

Sector definitions in the conventions above. The full set of exhibits within each sector is in the companion document *WP_sectors*.

Table 32: The four sectors: total and MNE exports, shares, leading parents

Sector	Total exports (\$bn)	MNE exports (\$bn)	Foreign (%)	Domestic (%)	Leading parents (share of foreign)
Agriculture	788.4	375.3	38.1	9.5	USA 0.24, CHE 0.11, NLD 0.10
Mining & fuels	810.3	597.2	49.2	24.5	GBR 0.36, USA 0.21, CAN 0.16
Manufacturing	788.3	470.1	51.6	8.0	USA 0.25, GBR 0.14, DEU 0.09
Rest	6.8	5.2	47.9	28.0	USA 0.39, GBR 0.12, ESP 0.11
All	2,393.8	1,447.8	46.3	14.1	

Agriculture = HS 01–24; Mining & fuels = HS 25–27 and 71 (precious metals and stones); Manufacturing = HS 28–97 (excl. 71); Rest = HS 98–99, chapter 00 and unclassifiable codes. Services are not observable in customs merchandise data. Pooled 2006–2022, nine LAC origins. Foreign / Domestic = share of the row's export value; leading parents = the three largest parent countries among the row's foreign MNEs with their share of the row's foreign-MNE value.

**Figure 25:** Foreign and domestic MNE shares by sector.

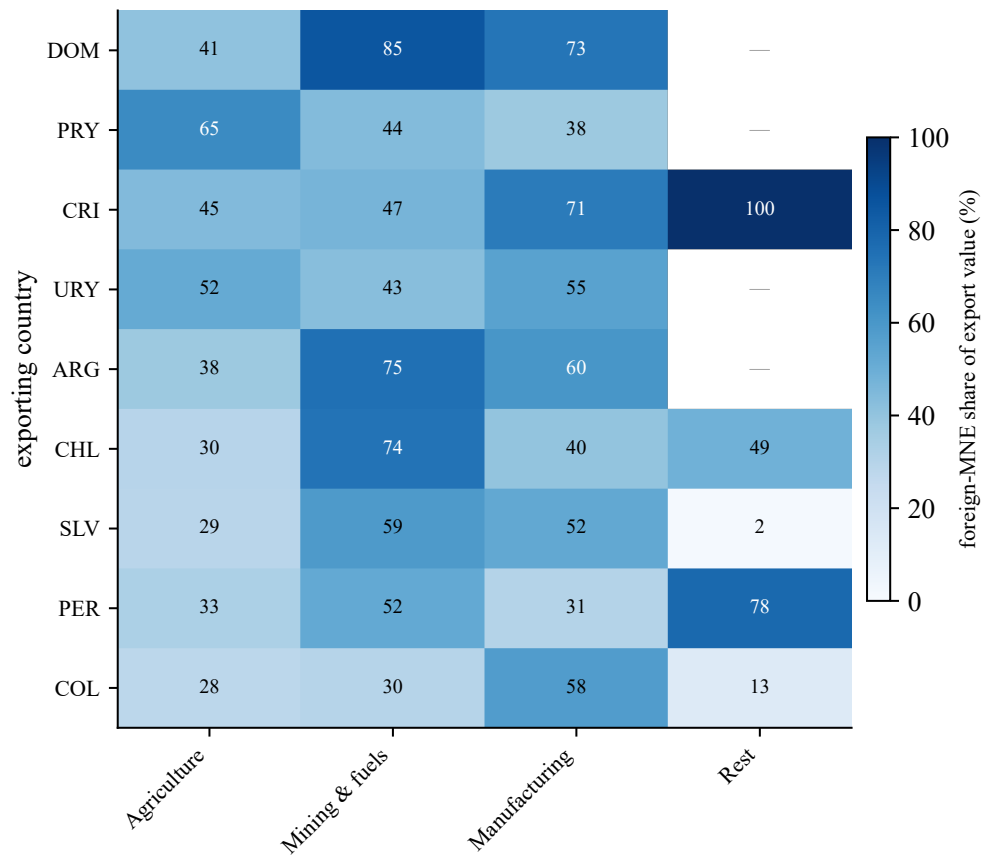


Figure 26: Foreign-MNE share by origin and sector (%).

Table 33: Composition (Panel A) and foreign-MNE share (Panel B) by origin and sector

Origin	Agriculture	Mining & fuels	Manufacturing	Rest	All
<i>Panel A: composition of the origin's exports in the scope, % (rows sum to 100)</i>					
ARG	57.4	9.9	32.7	0.0	100.0
CHL	21.9	29.7	46.9	1.4	100.0
COL	16.3	63.4	20.3	0.0	100.0
CRI	41.5	1.0	57.5	0.0	100.0
DOM	23.1	22.8	54.1	0.0	100.0
PER	19.4	61.0	19.6	0.0	100.0
PRY	85.2	1.1	13.7	0.0	100.0
SLV	22.1	3.0	74.9	0.0	100.0
URY	63.0	2.4	34.6	0.0	100.0
All origins	32.9	33.9	32.9	0.3	100.0
<i>Panel B: foreign-MNE share of export value within the cell, %</i>					
ARG	37.8	75.1	60.4	—	48.9
CHL	29.6	73.9	39.9	48.6	47.9
COL	28.0	30.1	57.8	12.7	35.3
CRI	44.5	46.9	70.9	100.0	59.7
DOM	40.7	85.3	73.2	—	68.4
PER	33.0	52.2	30.6	78.0	44.2
PRY	65.0	44.1	37.9	—	61.1
SLV	28.9	58.8	52.5	1.5	47.4
URY	51.8	42.9	55.3	—	52.8
All origins	38.1	49.2	51.6	47.9	46.3

Pooled 2006–2022. Panel A: share of each sub-sector in the origin's exports within the scope (rows sum to 100). Panel B: foreign-MNE share of export value in each origin × sub-sector cell ('—' = no exports); 'All' = the origin's overall foreign-MNE share in the scope.

Table 34: Foreign-MNE exports: destination region by sector (row %)

Sector / Destination region	Latin America	North America	Europe	Asia	Africa	Oceania	Rest of World	Total
Agriculture	31.1	13.8	20.2	28.1	6.1	0.4	0.4	300.6
Mining & fuels	14.3	14.6	24.7	45.5	0.5	0.2	0.2	398.3
Manufacturing	50.8	20.7	9.3	17.3	0.7	1.1	0.2	407.0
Rest	37.3	42.2	17.2	2.9	0.1	0.2	0.1	3.3

Foreign-MNE exports: destination mix of each sector, row percentages; last column = row total, USD bn.

Table 35: Memo: service-sector affiliates among matched exporters (NAICS of the affiliate)

NAICS sector of the matched exporting affiliate	% of matched export value
goods-producing or unknown	82.4
information, finance, professional, holdings (51–56)	7.7
wholesale / retail trade (42–45)	7.4
other services (61–92)	1.8
transport, logistics (48–49)	0.6

Memo. Customs data record goods; ‘services’ can only be approached through the affiliate’s own industry code. Matched exporters whose NAICS is a trade, logistics, finance or holding code export goods produced elsewhere in the group or bought from local suppliers.

A Numbers behind the figures

A.1 All goods

Table 36: Figure 1 numbers (foreign, domestic, total)

Origin	Foreign	Domestic	MNE total	Value (\$bn)
COL	0.353	0.391	0.744	528.0
DOM	0.684	0.013	0.697	68.8
CRI	0.597	0.036	0.633	104.7
URY	0.528	0.100	0.628	70.3
PRY	0.611	0.011	0.622	65.4
CHL	0.479	0.136	0.615	478.7
ARG	0.489	0.075	0.564	597.9
SLV	0.474	0.003	0.477	65.4
PER	0.442	0.023	0.466	414.5

Table 37: Figure 1 by parent, numbers

Origin	USA	GBR	CAN	LIE	DEU	NLD	CHE	JPN	PAN	BRA	Other	Dom.	MNE total	Value (\$bn)
COL	0.116	0.052	0.012	0.000	0.013	0.002	0.002	0.004	0.010	0.004	0.139	0.391	0.744	528.0
DOM	0.170	0.030	0.161	0.003	0.041	0.000	0.004	0.007	0.005	0.015	0.249	0.013	0.697	68.8
CRI	0.303	0.021	0.001	0.000	0.025	0.015	0.019	0.023	0.023	0.000	0.166	0.036	0.633	104.7
URY	0.049	0.018	0.009	0.000	0.010	0.038	0.009	0.024	0.007	0.085	0.277	0.100	0.628	70.3
PRY	0.301	0.003	0.000	0.000	0.000	0.021	0.000	0.016	0.001	0.009	0.259	0.011	0.622	65.4
CHL	0.026	0.210	0.070	0.062	0.005	0.000	0.004	0.011	0.024	0.001	0.067	0.136	0.615	478.7
ARG	0.081	0.029	0.023	0.039	0.045	0.055	0.048	0.034	0.000	0.013	0.121	0.075	0.564	597.9
SLV	0.279	0.014	0.011	0.000	0.005	0.005	0.000	0.014	0.038	0.002	0.106	0.003	0.477	65.4
PER	0.078	0.112	0.035	0.004	0.002	0.001	0.005	0.002	0.028	0.026	0.148	0.023	0.466	414.5

Figure 1 split by parent country. Value-weighted; pooled 2006–2022; nine LAC origins (Ecuador excluded). The foreign-MNE bar is split into the 10 largest parent countries by foreign-MNE export value in the scope and ‘Other foreign’ (all remaining parents, including matched firms with no recorded parent country); domestic MNEs have a parent in the exporting country.

Table 38: Figure 2 by parent, numbers

PCI quintile	USA	GBR	CAN	LIE	DEU	NLD	CHE	JPN	PAN	BRA	Other	Dom.	MNE total	Value (\$bn)
Q1	0.111	0.130	0.058	0.032	0.002	0.010	0.007	0.001	0.008	0.008	0.093	0.181	0.641	1,176.3
Q2	0.072	0.074	0.014	0.031	0.006	0.027	0.046	0.008	0.019	0.021	0.125	0.098	0.540	450.3
Q3	0.183	0.027	0.005	0.000	0.042	0.009	0.004	0.017	0.069	0.017	0.186	0.129	0.689	147.6
Q4	0.188	0.041	0.013	0.004	0.040	0.013	0.015	0.014	0.010	0.007	0.254	0.087	0.686	129.6
Q5	0.088	0.008	0.005	0.017	0.153	0.072	0.019	0.146	0.002	0.007	0.137	0.071	0.724	136.7

Figure 2 split by parent country. Quintiles of the Hausmann–Hidalgo Product Complexity Index over HS6 products. Value-weighted; pooled 2006–2022; nine LAC origins (Ecuador excluded). The foreign-MNE bar is split into the 10 largest parent countries by foreign-MNE export value in the scope and ‘Other foreign’ (all remaining parents, including matched firms with no recorded parent country); domestic MNEs have a parent in the exporting country.

Table 39: Figure 3 by parent, numbers

Technology category	USA	GBR	CAN	LIE	DEU	NLD	CHE	JPN	PAN	BRA	Other	Dom.	MNE total	Value (\$bn)
High tech manufacturing	0.220	0.012	0.003	0.000	0.034	0.005	0.023	0.011	0.006	0.009	0.210	0.080	0.613	60.4
Medium tech manufacturing	0.097	0.018	0.006	0.022	0.072	0.036	0.029	0.049	0.006	0.009	0.164	0.081	0.587	492.5
Low tech manufacturing	0.116	0.011	0.127	0.001	0.006	0.006	0.005	0.005	0.016	0.008	0.140	0.059	0.501	307.7
Primary and resource-based	0.094	0.128	0.025	0.030	0.003	0.013	0.013	0.004	0.018	0.014	0.114	0.182	0.638	1,436.4

Figure 3 split by parent country. Lall (2000) technology classification, four categories. Value-weighted; pooled 2006–2022; nine LAC origins (Ecuador excluded). The foreign-MNE bar is split into the 10 largest parent countries by foreign-MNE export value in the scope and ‘Other foreign’ (all remaining parents, including matched firms with no recorded parent country); domestic MNEs have a parent in the exporting country.